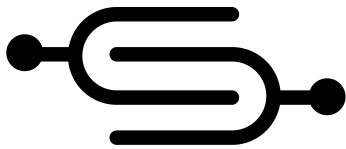


SOFTWARE ENGINEERING *notes*



SIGSOFT
SPECIAL INTEREST GROUP ON SOFTWARE ENGINEERING

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SIGSOFT EXECUTIVE COMMITTEE

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Chair: Thomas Zimmerman

Microsoft Research, Redmond, WA, USA,
<chair_SIGSOFT@acm.org>

Vice Chair: Marsha Chechik

University of Toronto, Canada,
<chechik@cs.toronto.edu>

AT-LARGE MEMBERS:

David Lo

Singapore Management University, Singapore,
<davidlo@smu.edu.sg>

Martin Robillard

McGill University, Canada, <martin@cs.mcgill.ca>

Lin Tan

Purdue University, USA, <lintan@purdue.edu>

Cristian Cadar

Imperial College London, UK,
<c.cadar@imperial.ac.uk>

Most Recent Past Chair: Nenad Medvidovic

University of Southern California, Los Angeles, CA,
USA, <nenod@usc.edu>

ACM PROGRAM COORDINATOR FOR SIGSOFT:

James Peeples

<peeples@hq.acm.org>

LIAISON MEMBERS:

Industry Liaison: Ahmed E. Hassan

Queen's University, <Ahmed@cs.queensu.ca>

International Liaison: Shing-Chi Cheung

The Hong Kong University of Science &
Technology, Hong Kong, <sccheung@cse.ust.hk>

iSoft Liaison: Srinivas Padmanabhuni

Chief Mentor, Tarah.ai,
<<https://www.linkedin.com/in/spadmanabhuni>
srinivas@tarahtech.com>

cSoft Liaison: Xiaoxing Ma

Nanjing University, China, <xxm@nju.edu.cn>

History Liaison: Tao Xie

University of Illinois at Urbana-Champaign, Urbana,
IL, USA, <taoxie@illinois.edu>

NON-EC POSITIONS

SEN Editor: Jacopo Soldani

University of Pisa, Italy, <jacopo.soldani@unipi.it>

SEN Production Editor: Ajay Jain

Adobe Systems, <ajjain@adobe.com>

Information Director: Raula Gaikovina Kula

Nara Institute of Science and Technology, Japan,
<raulak@is.naist.jp>

Digital Learning Co-Chairs: Shane McIntosh,

McGill University, <Shane.mcintosh@mcgill.ca>

Venera Arnaudova, Washington State University,
<venera.arnaudova@wsu.edu>

CAPS Coordinator: Nan Niu

University of Cincinnati, <niunn@ucmail.uc.edu>

Software Engineering Notes is an informal publication of the ACM Special Interest Group on Software Engineering (SIGSOFT) concerned with the cost-effective, timely development and maintenance of high-quality software. Relevant topics include requirements, specification, design and implementation methods, software maintenance, reuse, and re-engineering, quality assurance, measurement and evaluation, software processes, automated tools, and practical experience, etc. SIGSOFT seeks to address research and development issues in these areas and to provide a common ground for both, through sponsorship of conferences, symposia and workshops, the publication of SEN, and the dissemination of information via the SIGSOFT member email distribution list.

SIGSOFT sponsors an annual conference, Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), in the Fall and co-sponsors the International Conference of Software Engineering (ICSE) in the Spring, in addition to sponsoring several workshops and symposia each year on specialized and timely topics.

Material for publication in SEN should be prepared by using the SEN style (<https://www.sigsoft.org/SEN/subpub.html>). It should then be submitted via email to editors_sigsoft@acm.org, by sending their sources and/or .pdfs with all fonts embedded.

Contributions can be *articles*, *reports*, *reviews*, *columns*, *announcements*, or *book reviews*. The submission of *articles* is restricted to scientific/research articles that are part of a workshop for which SEN has agreed to be the publisher. As *columns*, instead, SEN welcomes contributions that discuss topics of interest among SIGSOFT members and the software engineering community at large. Columns should normally not be longer than 4 pages (exceptions may be allowed on a case-by-case basis) and trigger further discussion among community members. Finally, as *reports*, SEN welcomes summary reports from workshops conducted under the umbrella of ASE, ESEC/FSE, ICSE, and other major SIGSOFT sponsored conferences.

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Welcome Letter from the SIGSOFT Chair

Marsha Chechik
chechik@cs.toronto.edu

Dear SIGSOFT Community,

It is with great pleasure and excitement that I welcome you to the 2024–2027 term of the SIGSOFT Executive Committee. I am honoured to introduce our distinguished team:

- **Massimiliano Di Penta**, University of Sannio, Italy (Vice Chair)
- **David Lo**, Singapore Management University, Singapore (Secretary-Treasurer)
- **Emerson Murphy-Hill**, Microsoft, USA
- **Kelly Blincoe**, University of Auckland, New Zealand
- **Claire Le Goues**, Carnegie Mellon University, USA
- **Tom Zimmermann**, Microsoft Research, USA (Most Recent Past Chair)
- **Shakira Toomer**, ACM Support

Each member brings unique strengths and responsibilities to the committee, and I would like to outline their portfolios:

- **Massimiliano Di Penta**, as Vice Chair, will assist me in general duties and oversee SIGSOFT conference operations, including sponsored, co-sponsored, and in-cooperation events. This encompasses conference timelines, budgets, and publications.
- **Claire Le Goues** will manage SIGSOFT awards, ensuring that outstanding contributions in our field are recognized. She will also explore opportunities for new awards.
- **Emerson Murphy-Hill** will lead SIGSOFT's communications efforts, coordinating *Software Engineering Notes*, managing the SIGSOFT blog, and overseeing our social media and online presence.
- **David Lo** will focus on student mentorship and support, managing CAPS (Conference Attendance Program for Students) and other travel funds, summer schools, and special projects.
- **Kelly Blincoe** will spearhead membership growth and community outreach, ensuring that SIGSOFT remains an inclusive and vibrant community.

I would also like to take this opportunity to thank the members of the 2021–2024 SIGSOFT Executive Committee for their dedication and service to our community:

- **Lin Tan**, Purdue University, USA (Secretary-Treasurer)
- **Cristian Cadar**, Imperial College, UK
- **Martin Robillard**, McGill University, Canada
- And of course, **Tom Zimmermann**, who has served as Chair of the SIGSOFT Executive Committee for the past six years.

Both **David Lo** and I are continuing our work on the SIGSOFT Executive Committee, and we are excited about the future of SIGSOFT.

ANNOUNCEMENTS

Awards: SIGSOFT is now accepting nominations for our annual international awards. For more details, please visit <https://www2.sigsoft.org/awards/>.

- **Key Dates:**
 - **Deadline for submitting an intention to nominate (mandatory):** October 21, 2024
 - **Nomination deadline (firm):** November 4, 2024
 - **Notification of awards:** January 2025
 - **Award presentation:** at ICSE 2025

Open Positions: We are also looking for volunteers to support several SIGSOFT programs aimed at advancing the software engineering community. Open positions include:

- **Industry Chair:** Advocate for non-academic SIGSOFT members, contribute ideas for industry-oriented resources, and engage with the community.
- **Webinar Co-Chairs:** Identify and invite speakers to deliver webinars, interview them for the SIGSOFT blog and social media.
- **CAPS Co-Chairs:** Review student applications for travel funding to SIGSOFT-sponsored conferences.
- **Social Media Chair:** Promote SIGSOFT activities through social media channels.
- **Information Director:** Maintain SIGSOFT's digital assets (website, Mastodon server, mailing lists).
- **SIGSOFT Blog Editor:** Solicit and publish content of interest to the community.
- **Awards Committee Member:** Help select recipients of SIGSOFT awards.
- **Summer School Evaluation Committee Member:** Evaluate applications for SIGSOFT-funded summer schools.

If you're interested in volunteering, please complete the form at [this link](#).

I look forward to working closely with the new Executive Committee and with each of you to make a lasting impact on the SIGSOFT community.

Warm regards,

Marsha Chechik
 Chair, SIGSOFT Executive Committee (2024–2027)

Letter from the Editor

Jacopo Soldani
jacopo.soldani@unipi.it

Dear SIGSOFT-ers,

Welcome to the fourth issue of the 49th volume of our *Software Engineering Notes*, the first after the election of the new SIGSOFT Executive Committee. Let me then first congratulate with the elected committee members and with the chair, Marsha Checkik. I

am honoured to cooperate with you in delivering SIGSOFT mission! Let me also personally thank Tom Zimmerman and the former committee members, for the fruitful cooperation and very nice interactions. It was a pleasure to work with you!

Now, let's go back to introducing this issue, which – as usual – keeps long lasting traditions and tries to open new possible lines of discussion. As for our traditions, we keep posting on our long-lasting columns, i.e., the regular lists of ACM DL most downloaded and most cited papers (by Will Tracz), *Passages* (by Alex Groce), ACM risks (by Peter Neumann), and reflections on software industry (by Ahmed El-Deeb). As for possible new lines of discussion, instead, we are welcoming a contribution by Robin Nunkesser, who will provide his opinion on the end of mobile software engineering as we know it.

We also welcome reports from five workshops co-located with two flagship conferences for the SIGSOFT community, i.e., ICSE 2024 and FSE 2024. The reports indeed come from this year's editions of the (i) 5th international workshop on engineering and cybersecurity of critical systems and 2nd international workshop on software vulnerability management, (ii) 7th international workshop on software-intensive business, (iii) 1st international workshop on quantum software engineering: the next evolution, (iv) 6th international workshop on machine learning techniques for software quality evaluation, and (v) 2030 software engineering roadmap workshop.

As usual, let me thank the authors and the editorial board who participated to this issue of SEN, whose contributions were crucial to build this issue. Let me also thank you, readers!

Talk to you soon, and take care,



Jacopo Soldani, jacopo.soldani@unipi.it

PS: For any feedback on this issue, or suggestions to improve/open discussions on SEN, please feel free to contact us by writing to editors_sigsoft@acm.org (or to my personal email).

SIGSOFT FY'24 Annual Report

Thomas Zimmermann
zimmer@microsoft.com

The ACM Special Interest Group on Software Engineering (SIGSOFT) seeks to improve our ability to engineer software by stimulating interaction among practitioners, researchers, and educators; by fostering the professional development of software engineers; and by representing software engineers to professional, legal, and political entities.

This report provides a summary of key SIGSOFT activities over the past fiscal year. The software engineering community had a strong technical year with many significant contributions. SIGSOFT remains healthy both in terms of membership and finances. SIGSOFT sponsors 10+ conferences each year with several thousand attendees. To support its community SIGSOFT has invested in special projects, non-conference programs, as well as diversity, equity, and inclusion efforts.

EFFORTS RELATED TO DIVERSITY, EQUITY, AND INCLUSION

Several events or programs are focused on broadened participation either geographically, or among under-represented members of our community.

- SIGSOFT has dedicated **Liaisons for Diversity, Equity, and Inclusion** (DEI) who lead the DEI efforts for SIGSOFT and its conferences, but also connect and lead in the broader ACM community around Diversity, Equity, and Inclusion. The current liaisons are Jo Atlee, Kelly Blincoe and Alexander Serebrenik.
- As part of SIGSOFT's ongoing commitment to foster global educational opportunities and inclusivity, we launched in 2023 a **financial support program for summer/winter schools** focused on all aspects of software engineering across the world. The 2024 call for schools received 18 applications, of which we supported 13, with amounts of up to US\$ 10K. This program was coordinated by Marsha Chechik.
- The **CARES committee** is a resource comprising of well-known and respected people in the software engineering community who are approachable and willing to listen to and help people who experience or witness discrimination, harassment, or other ethical policy violations, either at SIGSOFT events or related to ACM publications. The current CARES chairs are Jo Atlee, Kelly Blincoe, Federica Sarro and Alexander Serebrenik
- SIGSOFT supports a **Student Mentorship Workshop (SMEW)** at the ICSE 2024 conference. This support was coordinated by Lin Tan.
- SIGSOFT supports efforts to bring students from Africa to its conferences. This effort was initiated and led by Yixue Xhao. For more details, please see the report on this initiative.¹ One student from Senegal successfully attended ICSE, we hope that this number can grow significantly in the past.
- SIGSOFT also financially supports the CRA-WP Graduate Cohort Workshops.

These efforts are in addition to ongoing exiting efforts related to diversity, equity, and inclusion.

- The major SIGSOFT conferences have EDI chairs, and there are guidelines on the kinds of activities the chairs can initiate and run. Most conferences also have a Code of Conduct and a Conference Diversity and Inclusion Plan.
- SIGSOFT has a larger community of people who are regularly thinking about EDI subjects: EDI conference chairs,

¹ From Zero to One: Lessons Learned on Bringing African Students to International Conferences. <https://medium.com/@yixue-zhao/from-zero-to-one-lessons-learned-on-bringing-african-students-to-international-conferences-74623059c4d4>

[to-one-lessons-learned-on-bringing-african-students-to-international-conferences-74623059c4d4](https://medium.com/@yixue-zhao/from-zero-to-one-lessons-learned-on-bringing-african-students-to-international-conferences-74623059c4d4)

participants of diversity panels, conference participants (when subscribing to Code of Conduct or selecting pronouns), etc.

- SIGSOFT has a dedicated travel support and conference attendance program called CAPS for members of underrepresented groups, students, or anyone who cannot afford the registration fees of a conference.
- To broaden the reach and membership, SIGSOFT has established national chapters in India (iSoft) and China (cSoft). Each chapter has a liaison member on the SIGSOFT EC.

AWARDS AND RECIPIENTS

The SIGSOFT awards program continues to recognize the many achievements of the software engineering community.

- The ACM SIGSOFT **Outstanding Research Award** was presented to **Tsong Yueh Chen** (Swinburne University of Technology) *“for contributions to software testing through the invention and development of metamorphic testing.”*
- The ACM SIGSOFT **Distinguished Service Award** was presented to **Massimiliano Di Penta** (University of Sannio) *“for outstanding service to the software engineering community, including serving as program co-chair for all premier software engineering conferences and several other events.”*
- The ACM SIGSOFT **Influential Educator Award** was presented to **Martin Robillard** (McGill University) *“for significant contributions to hands-on software design education, including a textbook and a learner-focused software modeling tool.”*
- The ACM SIGSOFT **Early Career Researcher Award** was presented to **Kevin Moran** (University of Central Florida) *“for outstanding contributions to improving the quality of mobile applications, through software testing, deep learning, and user interface development.”*
- The ACM SIGSOFT **Outstanding Dissertation Award** was presented to **Bianca Trinkenreich** (now at Oregon State University) for her Ph.D. thesis *“Understanding and Supporting Women’s Participation in Open Source Software”* (Northern Arizona University, advised by Igor Steinmacher and Marco Aurelio Gerosa).
- The ACM SIGBED/SIGSOFT Frank Anger Memorial Award was presented to **Max Camillo Eisele** (Saarland University and Bosch). Max Eisele is currently carrying out his PhD in the realm of security testing and fuzzing at Bosch Research, Germany.
- The ACM SIGSOFT **Impact Paper Award** recognizes a paper with exceptional impact on research or practice that was published in a SIGSOFT conference at least 11 years earlier. This year, the award went to the paper **“Learning from Examples to Improve Code Completion Systems”** (published at ESEC/FSE 2009) by **Marcel Bruch** (JetBrains), **Martin Monperrus** (KTH Royal Institute of Technology), and **Mira Mezini** (TU Darmstadt) *“for pioneering the use of machine learning to assist developers in generating code snippets, paving the way for today’s intelligent code assistance tools.”*

In total, the selection committees considered 50 nominations (10 for outstanding research, 5 for distinguished service, 4 for influential educator, 10 for impact paper, 11 for early career, 9 for

outstanding dissertation, and 1 for Frank Anger memorial), which is close to 30% more than last year. The decisions were made through a combination of online discussions and virtual meetings. Huge thanks to the SIGSOFT Award Committee Chairs (Walid Maalej, Massimiliano Di Penta, and Myra B. Cohen) and Deputy Chairs (Filippo Lanubile and Gordon Fraser), as well as the members of the respective committees, and all nominators. The SIGSOFT Awards Program is coordinated by David Lo.

In addition to the Impact Paper Award, many SIGSOFT conferences also have **Most Influential Paper Awards**, also known as “test of time awards”, which are given to papers that have appeared at a particular conference.

- At ICSE the SIGSOFT co-sponsored conference with the longest track record of awarding Most Influential Papers, the award went to the ICSE 2014 paper **“Coverage is not strongly correlated with test suite effectiveness”**, by **Laura Inozemtseva** and **Reid Holmes**.
- The ESEC/FSE conference (now called the FSE conference) recognized the ESEC/FSE 2013 paper **“Dynodroid: an input generation system for Android apps”** by **Arvind Machiry**, **Rohan Tahirani**, and **Mayur Naik**.

Many of SIGSOFT’s sponsored meetings also presented **Distinguished Paper Awards**. SIGSOFT allows up to 10% of the accepted papers to be selected for this award. The list of recognized papers is available at <http://sigsoft.org/awards/distinguishedPaperAward.html>

SIGNIFICANT PAPERS ON NEW AREAS

Software plays a prominent role in different application domains as well as other research areas in computer science, such as big data, human-computer interaction, mobile computing, artificial intelligence and machine learning, distributed systems, and security. Everything depends on software today and will continue in the future, even with the rise of large language models (LLMs). Over the past year, we have seen the rapid rise of these models in software systems. AI is also more frequently used to automate software engineering activities.

As examples for the breadth of the software engineering field, we highlight below distinguished papers from the ESEC/FSE 2023 and ICSE 2024, two of the main general software engineering conferences sponsored by SIGSOFT. The papers have been grouped and summarized into themes using ChatGPT. It will be exciting to see AI techniques accelerate science in the future.

Empirical studies of software development explored the impact of various interventions and external factors, such as interruptions, neurological influences, gender-based methodologies, and educational approaches, on software engineering practices and outcomes.

- **Breaking the Flow: A Study of Interruptions During Software Engineering Activities.** Yimeng Ma, Yu Huang, Kevin Leach (ICSE 2024)
- **Causal Relationships and Programming Outcomes: A Transcranial Magnetic Stimulation Experiment.** Hammad Ahmad,

Madeline Endres, Kaia Newman, Priscila Santiesteban, Emma Shedden, Westley Weimer (ICSE 2024)

- GenderMag Improves Discoverability in the Field, Especially for Women. Emerson Murphy-Hill, Alberto Elizondo, Ambar Murillo, Marian Harbach, Bogdan Vasilescu, Delphine Carlson, Florian Dessloch (ICSE 2024)
- A Longitudinal Study of Student Contributions to OSS vs. OSS4SG with a Lightweight Intervention. Z. Fang, M. Endres, T. Zimmermann, D. Ford, W. Weimer, K. Leach, Y. Huang (ESEC/FSE 2023)

Papers on **advanced analysis techniques** focused on automating and optimizing software development processes, including proof generation, SMT solving, local variable extraction, timing analysis, satisfiability checking, and property-based testing.

- Baldur: Whole-Proof Generation and Repair with Large Language Models. E. First, M. Rabe, T. Ringer, Y. Brun (ESEC/FSE 2023)
- Speeding up SMT Solving via Compiler Optimization. B. Mikek, Q. Zhang (ESEC/FSE 2023)
- An Automated Approach to Extracting Local Variables. X. Chi, H. Liu, G. Li, W. Wang, Y. Xia, Y. Jiang, Y. Zhang, W. Ji (ESEC/FSE 2023)
- A Highly Scalable, Hybrid, Cross-Platform Timing Analysis Framework Providing Accurate Differential Throughput Estimation via Instruction-Level Tracing. M. Hsu, F. Hetzelt, D. Gens, M. Maitland, M. Franz (ESEC/FSE 2023)
- Analyzing and Debugging Normative Requirements via Satisfiability Checking. Nick Feng, Lina Marsso, Sinem Getir Yaman, Yesugen Baatartogtokh, Reem Ayad, Victória Oldemburgo de Mello, Beverley Townsend, Isobel Standen, Ioannis Stefanakos, Calum Imrie, Genaína Nunes Rodrigues, Ana Cavalcanti, Radu Calinescu, Marsha Chechik. (ICSE 2024)
- Property-based testing in practice. Harrison Goldstein, Joseph W. Cutler, Daniel Dickstein, Benjamin C. Pierce, Andrew Head. (ICSE 2024)

Papers on **security in software systems** focused on enhancing the security and robustness of software systems through techniques such as API fuzzing, static vulnerability detection, monitoring data usage, and explainability-guided testing frameworks.

- EDEFuzz: A Web API Fuzzer for Excessive Data Exposures. Lianglu Pan, Shaanan Cohney, Toby Murray, Thuan Pham (ICSE 2024)
- Semantic-Enhanced Static Vulnerability Detection in Baseband Firmware. Yiming Liu, Cen Zhang, Feng Li, Yeting Li, Jianhua Zhou, Jian Wang, Lanlan Zhan, Yang Liu, Wei Huo (ICSE 2024)
- Attention! Your Copied Data is Under Monitoring: A Systematic Study of Clipboard Usage in Android Apps. Yongliang Chen, Ruqin Tang, Chaoshun Zuo, Xiaokuan Zhang, Lei Xue, Xiapu Luo, Qingchuan Zhao (ICSE 2024)
- Mate! Are You Really Aware? An Explainability-Guided Testing Framework for Robustness of Malware Detectors. R. Sun, M. Xue, G. Tyson, T. Dong, S. Li, S. Wang, H. Zhu, S. Camtepe, S. Nepal (ESEC/FSE 2023)

Several distinguished papers were on **smart contracts**. A smart contract is a self-executing program on a blockchain that automatically enforces and executes agreed-upon terms when specific conditions are met. Distinguished papers on smart contracts focused on transaction race conditions, accounting errors, and protection against oracle deviations

- TransRacer: Function Dependence-Guided Transaction Race Detection for Smart Contracts. C. Ma, W. Song, J. Huang (ESEC/FSE 2023)
- Towards Finding Accounting Errors in Smart Contracts. Brian Zhang (ICSE 2024)
- Safeguarding DeFi Smart Contracts against Oracle Deviations. Xun Deng, Sidi Mohamed Beillahi, Cyrus Minwalla, Han Du, Andreas Veneris, Fan Long (ICSE 2024)

Papers on **code search and recommendation** included papers on enhanced code suggestions, translation of complex idioms, and API recommendation through knowledge graph embedding.

- Code Search is All You Need? Improving Code Suggestions with Code Search. Junkai Chen, Xing Hu, Zhenhao Li, Cuiyun Gao, Xin Xia, David Lo (ICSE 2024)
- Hard to Read and Understand Pythonic Idioms? DeIdiom and Explain Them in Non-Idiomatic Equivalent Code. Zejun Zhang, Zhenchang Xing, Dehai Zhao, Qinghua Lu, Xiwei (Sherry) Xu, Liming Zhu (ICSE 2024)
- Recommending Analogical APIs via Knowledge Graph Embedding. M. Liu, Y. Yang, Y. Lou, X. Peng, Z. Zhou, X. Du, T. Yang (ESEC/FSE 2023)

And finally, many papers were dedicated to **software engineering and AI**. This is a rapidly growing area in the software community. These papers explore advancements in optimizing and diversifying machine learning pipelines, neural network generation, AI system robustness, and modular training methods, as well as improving compiler representations through pre-training.

- Can Machine Learning Pipelines Be Better Configured? Y. Wang, Y. Wang, T. Zhang, Y. Yu, S. Cheung, H. Yu, Z. Zhu (ESEC/FSE 2023)
- NeuRI: Diversifying DNN Generation via Inductive Rule Inference. J. Liu, J. Peng, Y. Wang, L. Zhang (ESEC/FSE 2023)
- Benchmarking Robustness of AI-enabled Multi-sensor Fusion Systems: Challenges and Opportunities. X. Gao, Z. Wang, Y. Feng, L. Ma, Z. Chen, B. Xu (ESEC/FSE 2023)
- LExecutor: Learning-Guided Execution. B. Souza, M. Pradel (ESEC/FSE 2023)
- Modularizing while Training: A New Paradigm for Modularizing DNN Models. Binhang Qi, Hailong Sun, Hongyu Zhang, Ruobing Zhao, Xiang Gao (ICSE 2024)
- FAIR: Flow Type-Aware Pre-Training of Compiler Intermediate Representations. Changan Niu, Chuanyi Li, Vincent Ng, David Lo, Bin Luo (ICSE 2024)

CONFERENCE ACTIVITY

Over the past year SIGSOFT sponsored and co-sponsored 10+ conferences related to software engineering. Conferences returned to an in-person format and occasionally offered hybrid formats as

well. The following is a list of some of the SIGSOFT sponsored conferences. The SIGSOFT conference program is overseen by the SIGSOFT Vice Chair, Marsha Chechik.

- ASE: Automated Software Engineering Conference
- CAIN: International Conference on AI Engineering – Software Engineering for AI
- DEBS: International Conference on Distributed and Event-based Systems
- ESEC/FSE: Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering: held annually.
- ESEM: International Symposium on Empirical Software Engineering and Management
- ICPE: International Conference on Performance Engineering
- ICPC: International Conference on Program Comprehension
- ICSE: International Conference on Software Engineering
- ISSTA: International Symposium on Software Testing & Analysis
- MoDELS: International Conference on Model-Driven Engineering Languages and Systems
- MSR: International Conference on Mining Software Repositories
- SEAMS: International Symposium on Software Engineering for Adaptive and Self-Managing Systems
- SPLC: International Software Product Line Conference

Given the rapid growth of AI and software engineering, SIGSOFT now co-sponsors conferences dedicated to this intersection.

- CAIN 2024: International Conference on AI Engineering – Software Engineering for AI
- FORGE 2024: Special Event on AI Foundation Models and Software Engineering
- AIware 2024: International Conference on AI-powered Software – Software for all and by all

SIGSOFT launched the **Proceeding of the ACM on Software Engineering (PACMSE)**, with FSE 2024 as the inaugural conference. PACMSE is a gold open access journal. For the first two years of PACMSE, SIGSOFT will cover the open access fees when authors can't. Huge thanks to Luciano Baresi for driving the PACMSE initiative for FSE.

Through the **Conference Aid Program for Students (CAPS)**, SIGSOFT provided travel support to conferences for dozens of graduate and undergraduate student-members as well as support to defray the costs of childcare for all members of our community (faculty qualify as well). The support is available for anyone who fits into one or more of the following categories:

- participants who identify as being a member of an underrepresented racial or ethnic group(s) in the place they work,
- participants based in low income and lower middle-income countries as identified by the World Bank List of Economies,
- undergraduate students,
- SIGSOFT members (both student and professional), or
- members of the SE community who cannot afford the registration fee.

The CAPS program is coordinated by Nan Niu. Detailed statistics about the CAPS program including open CAPS applications can be found here:

⇒ <https://acmsigsoft.github.io/caps/capsmain/>

SIGSOFT continued the **Conference Surplus Reinvestment**, which was launched in FY19. The surplus reinvestment for Year N of a conference is 25% of the Conference Net from Year N-2 and 25% of the Conference Net from Year N-3.

SPECIAL PROJECTS AND NON-CONFERENCE PROGRAMS

SIGSOFT has several existing programs to aid and expand our membership.

SEWORLD is a mailing list provided as a service to the Software Engineering community. It is intended primarily for the dissemination of time-sensitive information relevant to the field of software engineering research such as call for papers/participation and position announcements. The mailing list is moderated by Jinqui Yang and Elena Sherman.

The **Software Engineering Notes** (<https://sigsoft.org/SEN/>) is distributed exclusively in electronic form. The newsletter is published quarterly and provides an update to the community about ongoing efforts as well as other reports. The current Editor-in-Chief is Jacopo Soldani.

In 2022, SIGSOFT launched the **SIGSOFT Blog** (<https://sigsoft.medium.com/>) under the leadership of Cristian Cadar as an additional way to enhance the communication between the SIGSOFT Executive Committee and the software engineering community, and to create a sense of community within SIGSOFT, and publicize software engineering and SIGSOFT activities to the outside world. SIGSOFT also has **Webinar series** managed by Venera Arnaoudova and Shane McIntosh, where people can present to the SIGSOFT community.

The software engineering community has many community-driven vehicles to **promote open science** such as dedicated tracks for replications and negative results at conferences, policies for open science current draft is at: <https://github.com/acmsigsoft/open-science-policies>, and the ROSE festival <https://github.com/researchart/rose>, and pre-registered studies (EMSE journal in partnership with several SIGSOFT conferences, ACM TOSEM).

Since March 2023, SIGSOFT had a **Mastodon server** exclusively for SIGSOFT members (social.sigsoft.org). This server is maintained by Martin Robillard.

The **web presence** <https://sigsoft.org/> is maintained through GitHub Pages and is designed for multiple people to contribute via pull requests and at the same time preserves a history of changes that are made to the webpage. The webpage is maintained by Raula Gaikovina Kula who also serves as SIGSOFT's Information Director.

KEY ISSUES IN THE NEXT 2-3 YEARS

While SIGSOFT is stable and strong, there are several challenges that SIGSOFT and the broader software engineering community will continue to face.

- ACM is expected to transition to a **gold open access model** within the next few years. This may have an impact on conference attendance and publication revenue for SIGSOFT that needs to be carefully understood. Many SIGSOFT conferences are co-sponsored and alternate publication with IEEE, which is on a different path to open access than ACM.
- Despite a large growth in the number of software engineers and attendance at SIGSOFT conferences, SIGSOFT's **membership numbers** have not been growing. Providing more benefits to SIGSOFT members, increased visibility of SIGSOFT at conferences, and a focus on practitioners and regions beyond the United States will be important steps; however, it is also important to revisit what it means to be a SIGSOFT member. Many people are part of the SIGSOFT community by attending, publishing, volunteering for SIGSOFT and are not yet member of SIGSOFT.

Top 8 Most Downloaded Articles

<https://dl.acm.org/sig/sigsoft>

(as of 30 August 2024)

1. **Requirements engineering: a roadmap:** May 2000 in ICSE '00: Proceedings of the Conference on The Future of Software Engineering, Bashar Nuseibeh and Steve M Easterbrook. Downloaded 25323 (24,627) times.
2. **Software development lifecycle models:** May 2010, in ACM SIGSOFT Software Engineering Notes, by Nayan B Ruparelia, Downloaded 18,998 (18,343) times.
3. **A view of 20th and 21st century software engineering:** May 2006 in ICSE '06: Proceedings of the 28th international conference on Software engineering. Barry Boehm. Downloaded 17,931 (17,673) times.
4. **Managing the development of large software systems: concepts and techniques:** March 1987, in ICSE '87: Proceedings of the 9th international conference on Software Engineering, by Walter W Royce. Downloaded 17,754 (16,080) times.
5. **A spiral model of software development and enhancement:** ACM SIGSOFT Software Engineering Notes by B. Boehm Volume 11, Issue 4, August 1986 pp 14–24. Downloaded 12,901 (11,969) times.
6. **Foundations for the study of software architecture:** October 1992, in ACM SIGSOFT Software Engineering Notes by Dewayne E Perry and Alexander L. Wolf. Downloaded 12,324 (11,823) times.
7. **Case studies on testing object-oriented programs:** October 1991, in TAV4: Proceedings of the symposium on Testing, analysis, and verification. Roongko Doong and Phyllis G Frankl. Downloaded 12,174 (12,131) times.
8. **Software maintenance and evolution: a roadmap:** May 2000, in ICSE'00: Proceedings of the 9th international conference on Software Engineering by Authors: Yuchi Tian, Kexin Pei, Suman Jana, Baishakhi Ray. Pp 73-87. Downloaded 11,989 (new) times.

Top 8 Most Cited Articles

1. **Foundations for the study of software architecture –** October 1992. In ACM SIGSOFT Software Engineering Notes by Dewayne E. Perry, Alexander L. Wolf. Cited 1,426 (1,416) times.
2. **CUTE: a concolic unit testing engine for C –** September 2005 in ESEC/FSE-13: Proceedings of the 10th European software engineering conference held jointly with 13th ACM SIGSOFT international symposium on Foundations of software engineering, by Koushik Sen, Darko Marinov, Gul Agha. Cited 1,337 (1,317) times.
3. **Patterns in property specifications for finite-state verification –** May 1999 in ICSE '99: Proceedings of the 21st international conference on Software engineering, by Matthew B. Dwyer, George S. Avrunin, James C. Corbett. Cited 1,047 (1,029) times.
4. **Requirements Engineering: a Roadmap –** May 1999 in ICSE '00: Proceedings of the Conference on The Future of Software Engineering, by Bashar Nuseibeh, and Steve M Easterbrook. Cited 1,006 (980) times.
5. **Interface Automata –** September 2001 in ESEC/FSE-9: Proceedings of the 8th European software engineering conference held jointly with 9th ACM SIGSOFT international symposium on Foundations of software engineering by Luca de Alfaro, Thomas A. Henzinger. Cited 893 (889) times.
6. **Empirical evaluation of the tarantula automatic fault-localization technique:** November 2005 in ASE '05: Proceedings of the 20th IEEE/ACM international Conference on Automated software engineering by James Arthur Jones and Mary Jean Harrold. Cited 871 (863) times.
7. **Understanding Mobility based on GPS data – 200.** In UbiComp'08 Proceedings of the 10th International Conference on Ubiquitous Computing, by Yu Zheng, Quannan Li, Yukun Chen, Xing Xie, and Wei-Ying Ma . Cited 20 (758) times.
8. **Dynamo: a transparent dynamic optimization system –** May 2000. In PLDI '00: Proceedings of the ACM SIGPLAN 2000 conference on Programming language design and implementation by Vasanth Bala, Evelyn Duesterwald, Sanjeev Banerjia. Cited 818 (813) times.

Passages

Alex Groce
Northern Arizona University
United States
agroce@gmail.com

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<http://doi.org/10.1145/3696117.3696118>

Peter Seibel's *Coders at Work: Reflections on the Craft of Programming* is 600-odd pages of extraordinary software engineering wisdom and insight. It is also self-contradictory; these pages often disagree with each other, and sometimes actually argue with each other. As Donald Knuth says, near the end of this book, "the key is usually to have a stereo view instead of a one-dimensional view."

This book, inspired by the famous Paris Review *Writers at Work* series, consists of interviews with 15 luminaries of software creativity: Jamie Zawinski, Brad Fitzpatrick, Douglas Crockford, Brendan Eich, Joshua Bloch, Joe Armstrong, Simon Peyton Jones, Peter Norvig, Guy Steele, Dan Ingalls, L. Peter Deutsch, Ken Thompson, Fran Allen, Bernie Cosell, and Donald Knuth. One of these subjects appeared in the previous *Passages* selection consisting of a series of interviews, Federico Biancuzzi and Shane Warden's *Masterminds of Programming: Conversations with the Creators of Major Programming Languages*. This isn't a problem; Simon Peyton Jones is well worth listening to twice.

For comparison, the first (of many) volumes of the Paris Review series interviewed: E. M. Forster, Frank O'Connor, Dorothy Parker, Nelson Algren, Francois Mauriac, James Thurber, William Faulkner, Robert Penn Warren, Georges Simenon, Françoise Sagan, William Styron, Thornton Wilder, Truman Capote, Angus Wilson, Joyce Cary, and Alberto Moravia. I think we have to give the nod to Malcolm Cowley's collection, here, though I believe the Nobel and Turing counts are equal, and I don't know if anyone really reads Algren or Cary any more (more's the shame in Cary's case; his "First Trilogy" is wonderful).

Each interview is different, and the emphasis slowly shifts from modern JavaScript and web issues to topics dating back to the very beginning of computer science. However, Seibel makes it easy to contrast the ideas of the interviewees by asking certain questions (Seibel calls them his "standard questions") of every, or almost every, subject.

These standard questions include: "How did you learn to program?", "Do you still enjoy programming?", "What's your preferred debugging

tool?", "Do you use assertions?", "Do you use formal proofs?", "How do you go about reading a piece of software you didn't write?", "Top-down or bottom-up?", "How do you hire people?", "In your teams, does one person own a piece of code, or does everyone share ownership?", and perhaps most interestingly for the big picture, "Do you think of yourself as a scientist or an engineer or an artist or a craftsman or something else?" Seibel also asks most interviewees if they have ever used Knuth's literate programming approach.

The repeated mention of literate programming, and the overall link to the Paris Review series, connects this book to a fundamental theme of *Passages*: code is like writing, and is in fact written as much for other humans to read as for computers to execute. Many of the writers, I mean coders, Seibel talks to discuss this fundamental point, and many have strong opinions about how to make code that is a pleasure to read.

The answers to the full set of common questions, as you would expect from the list of coders, vary widely. There are some interesting commonalities: nobody really proves code correct; many people fear and loathe C++ or C, or both, though Thompson offers up a gruff argument in refutation to the C-related complaints.

I don't intend to summarize the things that are said in this book; there are too many of them, and most importantly, the meat of the book really lies in the un-summarizable personal stories of how each of these luminaries became involved in computing, and some of what each one of them actually did. The list of questions above suggests the fundamental questions of software addressed here. The book covers larger issues of project management, program structure, and approaches to refactoring, as well, and it's safe to say that almost no important issue of software engineering is entirely neglected. This book is a treasure trove that should be on every software engineer's shelf. Seibel asks some of his subjects: "Do you still program *for fun*?" and the answers suggest one final important common opinion of these diverse folks: we're lucky to be doing this!

ACM Risks Forum Quarterly Summary

Peter G. Neumann
 SRI Computer Science Lab
 333 Ravenswood Avenue, Menlo Park CA 94025-3493
 neumann@CSL.sri.com
<https://www.CSL.sri.com/users/neumann>
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ABSTRACT

This is an annotated summary of the main items contributed to the ACM Risks Forum in the most recent quarter, organized categorically. References (R i j) to the online Risks Forum denote RISKS vol i number j. Cited RISKS items generally identify contributors and sources, together with URLs. Official RISKS archives are available at www.risks.org, with nice html formatting and search engine courtesy of Lindsay Marshall at Newcastle:

<http://catless.ncl.ac.uk/Risks/i.j.html>

(also ftp://www.sri.com/risks). *CACM* Inside Risks:

<https://www.csl.sri.com/users/neumann/insiderisks.html>

TOTAL-SYSTEM TRUSTWORTHINESS

System Trustworthiness encompasses requirements such as system safety, security, reliability, survivability, guaranteed real-time requirements, and assurance that these requirements are met. The items in this section typically require system requirements from multiple disciplines.

- More on Boeing fuselage panel blowout (R 34 42)
 - Boeing Starliner software: While NASA continues to decide whether the thrusters on the Boeing Starliner now docked to the International Space Station can be relied upon to return the two astronauts who rode it up to the ISS back to Earth, a new issue has apparently arisen: the current flight software on board Starliner cannot perform an automated undocking from the space station and re-entry into Earth's atmosphere. (ArsTechnica, R 34 40)
- From Ars Technica: At first blush, this seems absurd. After all, Boeing's Orbital Flight Test 2 mission in May 2022 was a fully automated test of the Starliner vehicle. During this mission, the spacecraft flew up to the space station without crew on board and then returned to Earth six days later. Although the 2022 flight test was completed by a different Starliner vehicle, it clearly demonstrated the ability of the program's flight software to autonomously dock and return to Earth. Boeing did not respond to a media query about why this capability was removed for the crew flight test.
- CrowdStrike IT outage affected 8.5 million Windows users DELTA Airlines seriously hit; most major airlines affected except for Jet Blue and Southwest. (R 34 36); CrowdStrike update crashed the world's computers. What comes next? (R 34 36) The MTA's Old Computer Technology Kept Going During the MS-related Outage (R 34 36); Cyber Criminals Seek to Exploit CrowdStrike Outage (R 34 36); Why no public outrage over CrowdStrike/Microsoft and ATT failures? (R 34 37); CrowdStrike references (R 34 37); Microsoft's Global Sprawl Under Fire After Historic Outage (R

34 37); CrowdStrike and fuzz testing (R 34 38-39); Windows resiliency: Best practices and the path forward (R 34 39)

- Microsoft Security Sieve (R 34 35)
- Multiple Flaws in Microsoft macOS Apps Unpatched Despite Potential Risks (R 34 42)
- Apparent cyberattack at Seattle airport causes Internet outages (R 34 43)
- German Cyber-Agency Wants Changes in Microsoft, CrowdStrike Products after Outage (R 34 41)
- FAA Proposes New Cybersecurity Standards For Aircraft (R 34 41)
- U.S. Senate Votes 88 to 2 to Boost Nuclear Power Based on "New" Reactor Technology That is Old and Less Safe (R 34 33)
- Power Outages at Port of Los Angeles costly and nontrivial to recover (R 34 41)
- High-end racing bikes are now vulnerable to hacking (R 34 41-42)
- Amazon forced to recall 400K products that could kill, electrocute people (R 34 39)
- GPS Interference Over Land a Recurring Problem for Transatlantic Flights (R 34 34)
- Safety-critical aircraft parts (R 34 34)
- Boeing and Failures (R 34 36)
- Boeing 737 Max fabrication changes (R 34 34)
- Software engineers, not astronauts, are the heroes of today's space industry (R 34 34,37)
- Teenager Accused of Derailing Train and Posting Crash Video Online (R 34 39)

Dead Tesla traps toddler in hot car, raises concerns about electric doors (CarScoops, R 34 33))

Adults can use manual door releases from inside dead electric vehicles but younger ones can't.

* A Tesla in Arizona died and in the process trapped a toddler in the hot car.

* Firefighters had to break the window of the vehicle to get the child out as quickly as possible.

* The incident highlights the danger to those who own vehicles with electronic door releases. ...

(More in (R 34 34))

- Lithium Battery Fire Traps Drivers in Sweltering Heat on 'California Highway (R 34 38)

- A \$100-billion plan with "70

- NYSE technical error sends stocks tumbling (R 34 29); NYSE says bizarre glitch that showed Berkshire Hathaway down 99.97% has been resolved (R 34 29)

- From the *It's Not a Glitch* Dept. – 737-800 barely completed a takeoff (R 34 30)

- Waymo issues software and mapping recall after robotaxi crashes into a telephone pole (R 34 31); Waymo gets green light for expansion into LA and beyond (R 34 32)

- Driver using FSD, staring at phone, misses police officer before slamming into patrol car in Orange County CA. (R 34 31)

- U.S. Senate Votes 88 to 2 to boost nuclear power based on "new" reactor technology that is old and less safe! (R 34 33)

- Risks of Miracle AI Weapons (R 34 29)

- Artists threaten to leave Instagram in droves over AI art training (R 34 29)

- New technique can automate data curation for self-supervised pre-training of AI datasets (R 34 29)

- AI Systems Are Learning to Lie and Deceive (R 34 30)

- Hamane's Ai Pin (R 34 30)

- Clearview AI Used Your Face. Now You May Get a Stake in the Company: Facial recognition start-up doesn't have the funds to settle a class-action lawsuit, so lawyers are proposing equity for those whose faces were scraped from the Internet. (R 34 31)

- How a New Jersey man was wrongly arrested through facial recognition tech now in use in Ontario (R 34 31,33)

- In AI we trust, part II: Wherein AI adjudicates every Supreme Court case (R 34 33)

- Amazon-Powered AI Cameras Used to Detect Emotions of Unwitting UK Train Passengers (R 34 33)

- AI Accelerates Software Development to Breakneck Speeds (R 34 35)

- More productive AI => Self-Poisoned Training = GIGO (R 34 33)

- Ask Google Search a simple question, and get an AI Overview "guess" that is totally wrong China's AI-Powered Sex Dolls Set To Revolutionise Intimacy (R 34 34)

- Steven Wilson Struggles To Hear That It's Not Him Singingxo

AI-Created Songs (R 34 37)

- Limitless AI (R 34 37)

- AI captions (R 34 37)

- Personal experience with LLM AI Bios (R 34 37)

- Robots sacked, screenings shut down: a new movement of Ludites is rising up against AI (R 34 38)

- AI May Save Us, or May Construct Viruses to Kill Us (R 34 38)

- Restrictions on AI training data (R 34 38)

- Virginia Congresswoman Jennifer Wexton used an AI voice clone to address Congress (R 34 38)

- Robots sacked, screenings shut down: a new movement of Ludites is rising up against AI (R 34 39)

- Generative AI Has a 'Shoplifting' Problem. This Startup CEO Has a Plan to Fix It (R 34 40)

- Kroger unveils AI-powered automatic price gouger (R 34 40-42)

- We're Entering an AI Price-Fixing Dystopia (R 34 40)

- AI Cheating Is Getting Worse (R 34 41)

- U.S. Government Wants You – Yes, You – to Hunt Down Generative AI Flaws (R 34 41)

- Silicon Valley Is Coming Out in Force Against an AI-Safety Bill (R 34 41)

- A Loophole in Digital Wallet Security (R 34 41)

- AI is an energy hog. It's a strain on the power grid (R 34 41)

- AI and stand-up comedy (R 34 41)

- How much more water and power does AI computing demand? Tech firms don't want you to know (R 34 42)

- Recent bot campaign backing Poilievre shows AI easily accessible for political messaging: report (R 34 43)

- Without Guardrails, Generative AI Can Harm Education (R 34 43)

Healthcare Problems

- An Object Lesson From Covid on How to Destroy Public Trust: Officials should have told us what they knew, or at least leveled with us about what they didn't know. Public health officials squandered our faith in them by not being transparent. (Zeynep Tufekci, R 34 30)

- Bird Flu Shows That the U.S. Learned All the Wrong Lessons from Covid: The growing indifference has affected those still worried about Covid – last year the CDC stopped a lot of its pandemic data collection, making some basic facts like total deaths from Covid-19 much harder to track. (R 34 40)

- Halliburton Hit by Apparent Cyberattack (R 34 41)
- London Hospitals Face Major Disruptions After Cyberattack: ransomware cyberattack on Synnovis significantly disruptive (R 34 29)
- Phishing attack hits Los Angeles County public health agency, jeopardizing 200,000-plus residents' personal info (R 34 31)
- A Catastrophic Hospital Hack Ends in a Leak of 300M Patient Records (R 34 33)
- Leaked documents reveal patient safety issue at Amazon's One Medical Such issues have increased since One Medical shifted care to a call center staffed by contractors, employees say. (R 34 31)
- California plans to enlist AI to translate healthcare information (R 34 33)
- Even Doctors Like Me Are Falling Into This Medical Bill Trap (R 34 33)
- Another major hospital hack (R 34 34)
- U.S. Gender Care Is Ignoring Science – the Cass Review (R 34 36); rebuttal (R 34 37); counter-rebuttal (R 34 38); Gender Dysphoria and the Cass Review - A Summary of a Discussion (R 34 39)
- New findings shed light on risks and benefits of integrating AI into medical decision-making (R 34 37)
- Unfixable Infections (R 34 40)
- ARRL hit with ransomware (R 34 42)
- Almost half of FDA-approved AI medical devices are not trained on real patient data (R 34 42)
- Unintended consequences of building population tracking for COVID; public semi-nudity (R 34 35)

SECURITY and PRIVACY

General Security Issues:

Cryptography and Related Issues:

- NIST announces post quantum encryption standards; no hurry to use them! (R 34 40-41)
- Toward a Code-Breaking Quantum Computer (R 34 42)

Cryptocurrencies and Blockchains:

- How Researchers Cracked an 11-Year-Old Password to a \$3 Million Crypto Wallet: Thanks to a flaw in a decade-old version of the RoboForm password manager, and a bit of luck, researchers were able to unearth the password to a crypto wallet containing a fortune. (R 34 29)
- Crypto fanatics flock to Trump, hoping to *make bitcoin great again*. (R 34 38)
- Beware Politicians' Newfound Love of Cryptocurrency (R 34 40)

Surveillance and Espionage:

- Digital surveillance and customer isolation are individualizing the prices we pay (R 34 29)
- Security Firm Discovers Remote Worker Is North Korean Hacker (R 34 38)
- New Israeli Spyware (R 34 38)

Vulnerabilities:

- Scientists Find Security Risk in RISC-V Open-Source Chip (R 34 30)
- Microsoft Refused to Fix Flaw Years Before SolarWinds Hack (R 34 31)
- Retired engineer discovers 55-year-old bug in Lunar Lander computer game code (R 34 31)
- PHP+Windows Vulnerability (R 34 30)
- Flaw in Hundreds of Millions of AMD Chips Allows Deep, Virtually Unfixable Infections (R 34 40)
- Apple Patches AirPods Bluetooth Vulnerability That Could Allow Eavesdropping (R 34 32)
- Passwords Weakened by Advancements in Computing Processing (R 34 33)
- 10 Billion Passwords Exposed in Largest Leak Ever (R 34 35)
- New OpenSSH Vulnerability Discovered: Potential Remote Code Execution Risk (R 34 35)
- New tool for creating exploits (R 34 35)
- ID verification service reportedly left credentials wide open for a year (R 34 34)
- Worldwide BSOD outage (R 34 37)
- Secure Boot is Completely Compromised (R 34 37)
- Hackers could create traffic jams thanks to flaw in traffic-light controller, researcher says (R 34 37)
- Secure Boot is completely broken on 200+ models from 5 big device makers (R 34 38)
- New Flaws in Sonos Smart Speakers Allow Hackers to Eavesdrop on Users (R 34 40)
- Leaked github token could have put the entire python infrastructure at risk (R 34 39)
- Revoked DigiCert Digital Certificates: 27% Not Yet Replaced (R 34 41)
- Which devices on your network are most vulnerable? (R 34 41)
- RFID cards could turn into a global security mess after discovery of hardware backdoor (R 34 43)

- Scammers dupe chemical company into wiring \$60 million (R 34 43)

Exploits:

- Crooks plant backdoor in software used by courtrooms around the world (R 34 31)
- New Wi-Fi Takeover Attack – All Windows Users Warned To Update Now (R 34 31)
- 30,000 Dealerships Down – Ransomware Outage Outrage no.2 at CDK Global (R 34 34)
- French Museum Network Hit by Ransomware Attack (R 34 40)
- Hackers reverse engineer Ticketmaster (R 34 35)
- Spy v spy v spy: Jamming home wifi's by crims & cops (R 34 38)
- Devastating ransomware attack shuts down L.A. County courts Proofpoint Email Routing Flaw Exploited to Send Millions of Spoofed Phishing Emails (R 34 38)
- DDoS Attacks Surge 46
- The nation's best hackers found vulnerabilities in voting machines – but no time to fix them (R 34 40)

Data Breaches, Leaks, Anonymity, and Other Privacy Issues

- 361 million stolen accounts leaked on Telegram added to HIBP: password-stealing malware, credential stuffing attacks, data breaches added to the Have I Been Pwned data breach notification (R 34 29)
- Tile/Life360 Breach: Millions of Users' Data at Risk: Location tracking service leaks PII, because – incompetence? (R 34 31)
- Microsoft's Jaime Teevan doubles down on Windows Recall's "privacy sh*t-show" (R 34 30-31); Why Microsoft, Google, and others screw up (R 34 31) Microsoft and Recall (R 34 31)
- Hacker selling AMD data breach dated June 2024. Europol involved (R 34 33)
- Rampant Identity Theft Is Taxing the IRS (R 34 34)
- Hacker Accesses Internal Tile Tool That Provides Location Data to Cops (R 34 33)
- ID Verification Service for TikTok, Uber, X Exposed Driver Licenses (R 34 34)
- Feds *finally* starting to take privacy records seriously (R 34 35)
- Hackers steal call records of 'nearly all' AT customers (R 34 38)
- Personal Data of 3 Billion People Stolen in Hack, Suit Says (R 34 39)
- The Long Arms of Terms of Service (R 34 41)

- Park'N Fly reveals data breach affecting 1 million customer files (R 34 42)

- Android malware steals payment card data using previously unseen technique (R 34 43)

Usability, Education, and Related Human Issues:

- Don't You Dare Call Me Without Texting First (R 34 29)
- iOS 18 cracks down on apps asking for full address-book access (R 34 31)
- AWS MFA/2FA Changes (R 34 33)
- Second Factor SMS: Worse Than Its Reputation (R 34 35)

Other Internet issues, including the IoT and Clouds:

- Oral-B bricking Alexa toothbrush is cautionary tale against buzzy tech (R 34 29)
- Google, Cloudflare & Cisco Will Poison DNS to Stop Piracy Block Circumvention (R 34 31)
- Cisco to Lay Off Thousands in Latest Round of Tech Cuts (R 34 40)
- Intel Will Fire 15,000 Workers (R 34 40)
- Don't Let Your Domain Name Become a crime site (R 34 39)
- After sustained attacks by the GOP, Stanford Internet Observatory is being shut down (R 34 31)
- ICANN Approves DNS Top-Level Domain for Intranets (R 34 40)
- Logic Gone Astray: A Security Analysis Framework for the Control Plane Protocols of 5G Basebands (R 34 40)
- Android Phones Sold with Hidden Insecure Feature (R 34 41)
- Local Networks Go Global When Domain Names Collide (R 34 42)

Legislation/Law Enforcement/financial/other issues:

- How a Samsung Washing Machine Chime Triggered a YouTube Copyright Fiasco (R 34 29)
- Drone photographer pleads guilty to Espionage Act charges (R 34 36)
- About Kid's Online Safety Act and age verification (R 34 39)
- World's biggest music labels sue over AI copyright (R 34 32)
- U.S. to open broad antitrust probe into AI giants (R 34 30)
- More productive AI => Self-Poisoned Training GIGO (R 34 35)
- Perfect Apple Supply Chain Bug – Millions of Apps at Risk of CocoaPods RCE Security Boulevard) When AI tells you to verify

(R 34 36)

- Generative AI and the law: A federal law should be passed making AI firms fully responsible for any and all content disseminated from their generative AI systems. Period. No exceptions. (R 34 31,33)
- OpenAI illegally barred staff from airing safety risks, whistleblowers say (R 34 36)
- When it comes to math, AI is dumb (R 34 37)
- Encultured: an AI doomer's video game startup pivots to medicine. It'll be fine. (R 34 37)
- Apple signs on to Biden's responsible AI guidelines (R 34 38)
- The Age of the Drone Police Is Here (R 34 31)
- U.S. Surgeon General declares gun violence a public health crisis (R 34 32)
- In 6-3 ruling based on lack of standing, Supreme Court rules government may continue fighting social media misinformation (R 34 32)
- Incoming *WashPost* editor tied to self-described thief who claimed role in his reporting (R 34 33)
- Supreme Court accidentally posts with Biden admin on Idaho abortion case with Biden admin on Idaho abortion case (R 34 34)
- Argentina will use AI to 'predict future crimes' but experts worry for citizens' rights (R 34 39)
- Feds sue Georgia Tech for lying bigly about computer security (R 34 41-43)
- Florida company faces multiple lawsuits after massive data breach (R 34 41)
- How Section 230 Is Being Used Against Tech Giants Like Meta (R 34 42)
- Telegram billionaire co-founder Pavel Durov arrested (R 34 42-43) • Foreign Policy: TikTok ban & global data commons (R 34 43)
- Two policy articles suggested by Dan Geer on due care (R 34 42-43)
- Telco fined \$1M for transmitting Biden deepfake without verifying Caller ID (R 34 43)

Disinformation, Lies, Fakes, Frauds, Biases, etc.

- Deepfake of U.S. Official Appears After Shift on Ukraine Attacks in Russia: A manufactured video fabricated comments by the State Department spokesman, Matthew Miller. (R 34 29)
- Fake News Reports and Videos Seek to Undermine the Paris Olympics: Russian propagandists are spoofing broadcasters and mimicking French and U.S. intelligence agencies to stoke fear about security at the Games. (R 34 29)

- They Spent Their Life Savings on Life Coaching (R 34 29)
- A reportedly fake group recruited a real candidate for Congress (R 34 31)
- Annandale man wins fraud case against a bank (R 34 30)
- Vax: Sometimes it seems these days an entire topic is fake, not to mention everything said about it. (R 34 31)
- Call to ban DJI drones introduced in U.S. Senate, company responds (R 34 40)
- Trolls Used Her Face to Make Fake Porn. There Was Nothing She Could Do. (R 34 39)
- How One Man Lost \$740,000 to Scammers Targeting His Retirement Savings (R 34 39)
- Microsoft Copilot makes a court reporter into a child molester (R 34 41)
- Regulators May Not Like Deepfakes, But Businesses Are Using Them Anyway (R 34 41)
- Fake QR codes posted on Redondo Beach parking meters to scam drivers, police say (R 34 42-43)

ELECTION INTEGRITY

- Musk calls for elimination of electronic voting machines, full hand count of all ballots (R 34 33); How Crypto Money Is Poised to Influence the Election (R 34 33); Voting machine contract under scrutiny following hundreds of discrepancies in Puerto Rico's primaries (R 34 31)
- Ding dong drama: Video doorbells have UK election campaigners spooked
- Electronic voting in France from abroad (R 34 35-36) (R 34 34)
- Voice assistants and AI chatbots still can't say who won the 2020 election (R 34 34)
- In GA the Biggest Election Breach in History Has Gone Uninvestigated (R 34 36)
- Illinois Voter Data Exposed by Unsecured Databases (R 34 40-41)
- AI Detection Tools Often Fail to Catch Election Deepfakes (R 34 41)
- Trump Campaign Confirms It Was Hacked (R 34 40)
- Trump posts fake AI images of Taylor Swift and Swifties, falsely suggesting he has the singer's support (R 34 41)
- U.S. and Allies Issue Rare Warning on Chinese Hacking Group (R 34 35)
- Russia Breaches TeamViewer: No Evidence Billions of Devices at Risk (R 34 35)
- Moscow's Spies Were Stealing U.S. Tech, Until the FBI

Started a Sabotage Campaign (R 34 43)

- Canada warns of AI-driven Russian ‘bot farm’ spreading disinformation online (R 34 35-36)
- Japan Runs on Vending Machines. It’s About to Break Millions of Them. Them with incompatible new Yen currency. (NYTimes) (R 34 31)
- A Pacific Island With Ties to Taiwan Was Hacked. Was It Political? (R 34 29)
- Dash to Ditch Paper Money in Sweden Created a Playground for Criminals (R 34 33)
- Antivirus Shuffle over Kaspersky (R 34 33-34)
- In Ukraine War, AI Begins Ushering In an Age of Killer Robots (R 34 36-37)
- UK PM Warns Social Media Firms After Misinformation Fuels Riots (R 34 40)
- Meta Kills Off Misinformation Tracking Tool (R 34 41)
- AT&T says hacker stole call records of ‘nearly all’ wireless customers (R 34 36)
- Data breach exposes millions of mSpy spyware customers (R 34 36)
- Rite Aid says June data breach impacts 2.2 million people (R 34 36)

MISCELLANEOUS ITEMS

- Harvard grad who went off script to address Gaza protests said she quietly revised her speech last minute (R 34 29); New Bone of Contention: Trigger Warnings in Archaeology Class: Professors from Harvard to Cambridge feel obliged to caution students before displaying ancient human remains; drawings and plastic skeletons fill the void. (R 34 29)
- On teaching Ethics (R 34 29)
- Study finds 268% higher failure rates for Agile software projects The best video I’ve seen explaining the technical reasons why keeping AM radios in cars is so important! (R 34 30)
- GM to Cut More Than 1,000 Software Engineers, Mostly in U.S. (R 34 41)
- Wells Fargo Fires Over a Dozen for ‘mouse jiggling’ (R 34 31)
- Colorado discovers error causing EV tax credit denials Architecture (R 34 30)
- The surprisingly not so doomed effort to force U.S. drivers to stop speeding (R 34 31); [See my comment on the driver crashing at 100 mph in RISKS-34.25-26 about having some sort of sanity limit. PGN]
- MXThunder and FBL (R 34 31)
- Mass.’s 911 system crashes... (R 34 33); Mass. 911 system back

online after outage (R 34 33)

- NYC traffic Congestion Pricing paused; but its LPR tracking not paused (R 34 33)
- Nations Warn Key Open-Source Programs Not Sufficiently Protected (R 34 35); Switzerland now requires all government software to be open-source (R 34 37-39)
- Ozone Hole Mk. II (R 34 33-35,37)
- ID verification service fails (R 34 34)
- Firefighter charity bot call (R 34 34-35)
- What to do when you send money to the wrong person through Zelle (R 34 34-35)
- The end of the world (R 34 34)
- Prominent Short Seller Made Millions Off Bait-and-Switch Scheme, U.S. Says (R 34 38)
- Google reverts TV YouTube app to original search history behavior (R 34 38-39)
- Chipmaking Giant Learns What Works in Taiwan Doesn’t in Arizona (R 34 40)
- Outages Plague Trading Platforms During Stock-Market Selloff (R 34 40)
- Canada’s food supply - still under threat? (R 34 40)
- Excess memes and ‘reply all’ mails are bad for climate, researcher warns (R 34 40)
- Experts to PNT leaders: “It’s not working!” (R 34 40)
- These ‘living computers’ are made from human neurons and you can rent one for \$500 a month (R 34 41)
- Number of Women Taking CS Degrees in the UK Continues to Grow (R 34 41)
- Is it safer to use an app or a website on your phone? (R 34 41)
- Nightly Waymo Robotaxi Parking Lot Honkfest Is Waking Neighbors (R 34 41)
- Denver Water’s loss of pressure at 5 AM every Monday is the same problem as San Francisco’s 4 AM robot taxi honkfest (R 34 41)
- Birmingham council faces huge loss over Oracle debacle (R 34 41,43)
- Policy, due care, and the failure of Heartland Tri-State Bank (R 34 41-42)
- Biometrics in the workplace may be the way of the future. But at what cost? (R 34 42)
- Apple to Let iPhone Users Delete Safari, Other Native Apps to Comply With EU Law (R 34 43)

Can I Trust you? Ethics in AI

Ahmed El-Deeb

Amazon

Krausenstr. 38, 10117 Berlin

+49 176 434 28719

ahmed.eldeeb@aucegypt.edu

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ABSTRACT

We all like to use and play with utilities and tools that solve daily problems to us. After we do play with it for sometime, we may pause and ask “hmm, but would that enable some bad guys to kill somebody?” Now we are being ethical about the usage of that tool. AI has been in the hands of consumers for the last couple of years, expanding the horizon to what users can do. But I am not sure if lay users have actually paused to question themselves whether they are breaking any ethics or they are so consumed getting things done with the impressive tool. AI is enabling us to be more productive; however, it carries with it severe ethical violations; right now not next month or next year. As I write this article, there are many unethical usages of AI being done without any notice. Think of those who publish short articles written not by them but by AI, those who faked photos, those who wrote their essays, or that one who authored a piece of poetry. This is on individual level, but such violations could be large scale when it comes to corporations. What about those who are the actual makers of AI – are they trustable or developing models that carry some bias? This article is aimed at exploring the topic of Ethics in AI to help us understand implications and path forward.

Categories and Subject Descriptors

A.0.3 [GENERAL]: Tech Industry and Development Teams.

General Terms

AI, Generative AI.

Keywords

Metrics, Measurements, Customer Satisfaction, Software Quality.

INTRODUCTION

Each time I read a piece of info or query my ChatGPT about a topic I can't help myself but think to what degree should I trust that. It's a beautiful tool, but is the content produced evenly or does that person who is publishing something utilizing AI in the mix whether they did their homework of scrutinizing their AI agent. What are the guardrails that are given to us with which we can be assured that what we are consumed out of AI is trustable and those who uses what's produced by AI are trustable as well.

KEY ETHICAL CONCERNS

Algorithmic Bias

Imagine an institution or a government uses AI to decide on public policy or filter out list of candidates only to find that algorithm disadvantages people of color or women. That's not a far-fetched scenario. The National Institute of Standards and Technology (NIST) has conducted a study on facial recognition algorithms from 100 developers over 189 organizations; such as, Toshiba, Intel, and Microsoft found that there are demographic differentials across the majority of algorithms [1]. Thus we need to be aware that AI could be biased and that would be a first step

toward scrutinizing its output or at least use it in an augmented way rather than an omni-knowledge tool.

Data Privacy

Data is another critical concern. The cornerstone of AI advancement is data so that we can train the model. That said, data entered by you is in fact getting into that machinery of training the AI model such that it is part of the entire collective intelligence of the model enabling it to answer questions or exchange with others too. That's all fine, but it is not in certain cases where data is shared with third party or exposed to unauthorized personnel, or how that data is stored away from feeding it in the model. This is not just restricted to AI tools you use with your own data, but also about whatever information is available online. For example, AI models can sniff CVs or posts readily online to use them as an input to train the model [2].

Transparency and Explainability

With heavily using AI comes the first question: should I trust you? This is a very critical question beyond the simple copy/paste of text. This even becomes more critical if you are using the output into something sensitive. To answer this question of whether you can trust that output from your AI agent is that you have to know how the model arrived at that outcome; what methods it used, which data, and what reasoning methodology. Unfortunately, interpretability and explainability is not something inherently available to you. The fact that AI output is not traceable is a severe trust issue [3].

CONCLUSION

AI tools are not heaven on Earth and have impactful role in our lives when it comes to information fed into it and used out of them. We may be rushing into using them, but what will bite us in the future is how can we trust the Ethics of AI when it comes to fairness, privacy, and trust. The fundamental factors to ensure is that we need AI to be fair, to be private, and to be transparent. These are the minimum ethical factors for us to use AI.

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Mobile Software Engineering is Coming to an End (Like All Good Things Must)

Robin Nunkesser
 Hamm-Lippstadt University of Applied Sciences
 Hamm, Germany
robin.nunkesser@hshl.de
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ABSTRACT

Mobile Software Engineering as a separate research field has been relevant to speeding up Software Engineering research in the area of mobile devices and applications. However, the existence of Mobile Software Engineering as a separate research field is coming to an end. This development has implications for existing and future research. These implications and resulting opportunities are discussed in this paper.

1. INTRODUCTION

There are sometimes moments in research when areas split off in order to develop separately more quickly. Sometimes we encounter situations whereby common ground in different research areas is discovered and, in the best case, researchers combine their strengths. This potential combination of strengths is also possible in the comparable situation whereby a research area splits off temporarily and is reunited later.

All of the above situations offer great opportunities for research. In this paper, the development of Mobile Software Engineering as a separate research field and the opportunities that arise from the end of this development are discussed.

Sometimes the opportunities are so great that they can be compared to a gold rush. I have already been lucky enough to participate in one such long gold rush originating from the interface between computer science and statistics [7]:

From the viewpoint of applied computational complexity, statistics is a gold mine, for it provides a rich and extensive source of unanalyzed algorithms and computational procedures.

However, the speed of current research is not comparable to past research phases. In this, as in so many other areas, the effects of exponential growth in recent years have surprised us, even though they were mathematically known (according to [1] this is humanity's greatest shortcoming). According to [2], research output is also growing exponentially. We therefore expect future "gold rushes" to become shorter and more intense, which is why we should be well prepared for them.

According to [5], the research field of Mobile Software Engineering slowly split off from Software Engineering in the early 2010s. The field has since developed rapidly and is now coming to an end (of the split-off). We should see the opportunities in this development and use them to our advantage. Maybe we can even participate in a little gold rush.

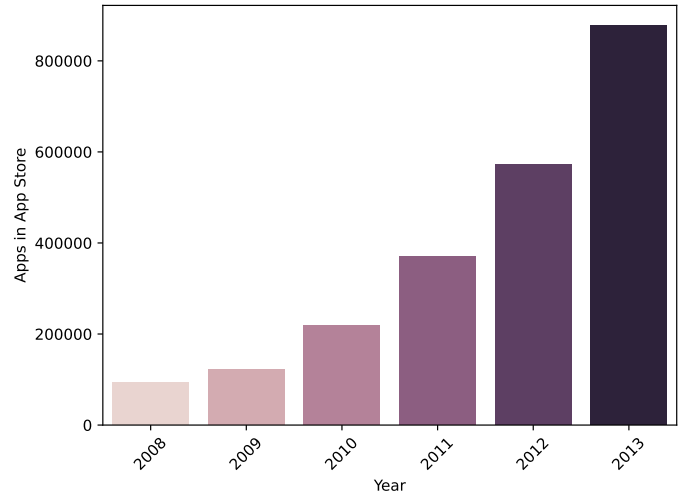


Figure 1: Exponential Growth of Distributed Apps (2008–2013)

2. MOBILE SOFTWARE ENGINEERING

Let us first trace the development of the research area of Mobile Software Engineering.

2.1 The Mobile Revolution

The development leading to separate research in Mobile Software Engineering started in 2008 with the introduction of Apple's App Store [5]. The App Store marked a starting point for exponential growth of the usage and distribution of mobile software only comparable to the growth of AI usage (in January 2023, only two months after its launch in late November 2022, ChatGPT had 100 million monthly active users, according to data from Similarweb). It was nothing short of a revolution (again only comparable to what we have seen with AI since 2023). It took only nine months to reach one billion app downloads in the App Store [5], which continued to increase exponentially (Figure 1 shows the number of active applications in the US App Store for the years 2008–2013).

2.2 First Research Agenda

In 2010, Wasserman [8] recognized the need for separate research:

Despite the development of 300,000+ mobile applications, there's still not much formal research around their engineering processes.

Wasserman formulated a research agenda, which [5] summarized as:

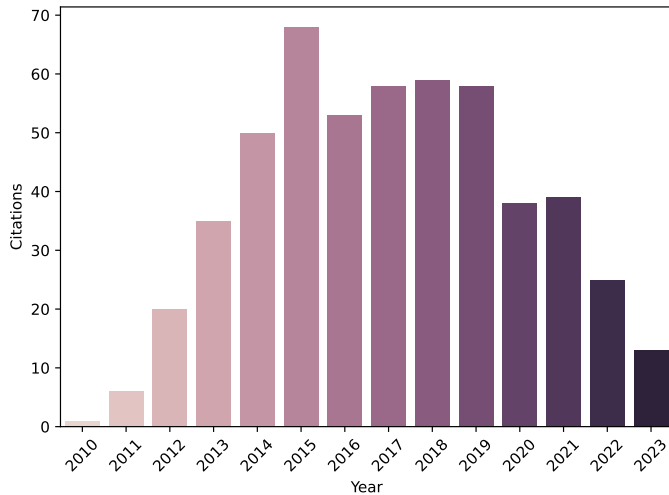


Figure 2: Citations of [8] by year according to ResearchGate (originally published in [5])

- User Experience
- Code reuse
- Networking
- Energy efficiency
- Data integrity
- Testing
- Portability and cross-platform development

2.3 Split-Off for Separate Research

In the years following the rise of iOS and Android devices and the work of [8], separate research in Mobile Software Engineering was successfully initiated. When we consider the International Conference on Software Engineering (ICSE) and the Symposium on the Foundations of Software Engineering (FSE), the two flagship software engineering conferences, we see that the need for a separate research field was recognized at ICSE with the International Conference on Mobile Software Engineering and Systems (MOBILESoft)¹, which was first held in 2014, and at FSE with the International Workshop on Software Development Lifecycle for Mobile (DeMobile)², which was first held in 2013.

2.4 All Good Things Must Come to an End

[5] states that “the need for separate research in Mobile Software Engineering is coming to an end.” This observation is based on figures such as the low number of accepted papers at MobileSOFT in recent years (e.g. only nine accepted papers in 2023) or the declining numbers of citations of Wasserman’s research agenda [8] (see Figure 2).

However, this goes hand in hand with a general rise in publications concerned with formerly mobile specific topics [5]. The research agenda proposed in [8] is now part of the research agenda of Software Engineering in its entirety. According to [5], only 11.64% of papers published in 2023 concerning topics connected

¹MOBILESoft is held in conjunction with ICSE

²DeMobile was held in conjunction with FSE in 2013 and 2014

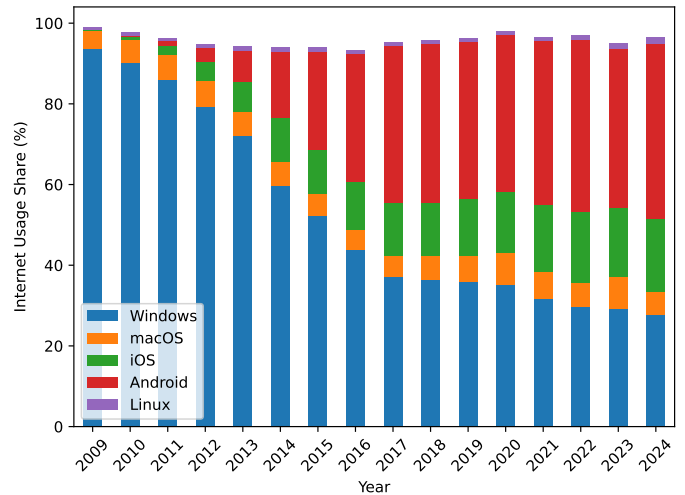


Figure 3: Internet Usage Share of Selected Operating Systems between 2009 and 2024 according to StatCounter Global Stats

with Wasserman’s research agenda also cover mobile-specific topics. Many topics that were originally mobile specific are now part of general Software Engineering research.

2.5 Mobile is Here to Stay

This is obviously not the end of mobile itself, just the end of the split-off of Mobile Software Engineering. Mobile devices and applications are here to stay and more relevant than ever.

According to StatCounter Global Stats, Android first exceeded Windows in terms of internet usage in 2017 and in 2023 Android had a market share of 39.56% compared to 29.21% for Windows (see Figure 3).

Most software targeting the consumer market should therefore support mobile devices or even be mobile first. The use of mobile devices is also increasing in the enterprise market (see e.g. [6] for examples from controlling). It is not an exaggeration to say that any future Software Engineering research should at least initially consider whether mobile is a target platform and what impact this will have.

3. IMPLICATIONS FOR RESEARCH

The developments described have implications for existing and future research.

3.1 Existing Research

In relation to existing research, we can distinguish between:

- Software Engineering Research that has not yet considered mobile devices and applications
- Mobile Software Engineering Research that has not yet considered the impact on Software Engineering in its entirety
- Research that has already considered the impact of mobile devices and applications on Software Engineering

For the first group, the implications are clear: there are opportunities in reconsidering research conducted before the split-off

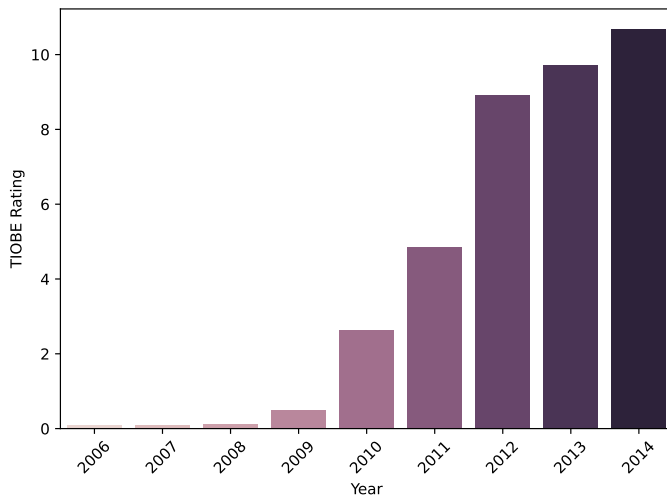


Figure 4: TIOBE Index of Objective-C between 2006 and 2014

of Mobile Software Engineering. In this way, results that were previously not known to be applicable to mobile devices and applications can be discovered. The implications are also clear for the second group: there may be results that are also applicable to Software Engineering in its entirety. There is nothing to do for the third group.

Are these opportunities great enough for a little gold rush? There are good reasons to be optimistic. The exponential growth of mobile devices and applications had some impact on research before and after the split-off of Mobile Software Engineering. There are at least two factors that have had a negative impact: the proportion of less-experienced developers and researchers was too high and the positioning between web development, embedded development, and general software engineering was unclear.

To prove the first factor, it is worth looking at the situation of programming for iOS. Since 2008, apps on iOS could be programmed using Objective-C, a language that previously had almost no significance outside of Apple. Figure 4 shows the rapid increase in the importance of this language. It can be assumed that this development has brought many inexperienced developers and developers without a solid foundation in software engineering into mobile development. The second factor has similar reasons: there were simply not enough experts for mobile applications to cover the demand and so implementers from different areas carried out the implementations. This led to problems that were comparable to the boom in web development which were described by [3] as:

The ad-hoc and chaotic manner in which the majority of web applications are developed is a real cause for concern.

The likely impact on research results is that there are still many classic software engineering insights that can be applied to mobile development (see [4] for an example). It is also likely that mobile-specific research often does not pay enough attention to generalization.

3.2 Future Research

Researchers who are currently working on mobile-specific topics should also consider the impact of their research on Software Engineering. Conversely, Software Engineering researchers who have not included mobile devices and applications in their research so far should consider the impact of mobile on their research. Any future new Software Engineering research should at least initially consider whether mobile is a target platform and what impact this will have.

4. SUMMARY

Mobile Software Engineering as a separate research field is coming to an end. This development has implications for existing and future research. Systematically looking at mobile-specific results and checking whether they are generalizable offers opportunities for research.

Other recommended actions for researchers are:

- If you are looking for software engineering results for mobile devices or apps, consider the applicability of non mobile-specific results.
- If you are researching supposedly mobile-specific topics, consider whether the results can also be useful for other areas.
- When starting new research, consider whether mobile devices make a difference to the research and mention whether they are included or excluded.

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Report on the 5th International Workshop on Engineering and Cybersecurity of Critical Systems and 2nd International Workshop on Software Vulnerability Management (EnCyCriS/SVM – 2024)

John Eidar Simensen

Institute for Energy Technology
Halden, Norway

John.eidar.simensen@ife.no

Coralie Esnoul

Institute for Energy Technology
Halden, Norway

coralie.esnoul@ife.no

Eunkyoung Jee

Korea Advanced Institute of
Science and Technology
Daejeon, Republic of Korea

ekjee@se.kaist.ac.kr

Ali Babar

CREST – Centre for Research on
Engineering Software Technologies,
University of Adelaide
Adelaide, Australia

ali.babar@adelaide.edu.au

Triet Huynh Minh Le

CREST – Centre for Research on
Engineering Software Technologies,
University of Adelaide
Adelaide, Australia

triet.h.le@adelaide.edu.au

Awais Rashid

University of Bristol
Bristol, England

awais.rashid@bristol.ac.uk

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ABSTRACT

Increasing system interconnectivity, decentralization, and introduction of new, more intelligent technologies, result in critical infrastructures becoming exposed to increased risk of cyber, physical, and combine cyber-physical attacks. Cyber-attacks on critical systems can inflict severe consequences on to people, society, economy, and national security, and can have adverse effects on safety and reliability of critical infrastructures. The joint EnCyCriS-SVM workshop facilitates discourse and discussion among researchers, practitioner, and students who are working on challenges and solutions related to the industrial revolution. Focus is given on sharing industry experience and project results pertaining to cyber threats on critical systems; secure software engineering; and attack detection and response mechanisms.

Keywords

Cybersecurity, Safety, Software engineering, Systems engineering, Critical infrastructures.

1. INTRODUCTION

Software permeates modern society. Within critical infrastructures and systems providing important societal services, there have been considerable digitization efforts the last decade. To address critical infrastructures vulnerabilities in design, development, implementation, operation and maintenance, a Joint Workshop is arranged between the International Workshop on Engineering and security of Critical Systems (EnCyCriS) and the International Workshop on Software Vulnerability Management (SVM).

An effect of the 4th industrial revolution is that cyber physical systems and software are in continuous growth in their complexity. Complexity of data, and system integration are becoming increasingly important for business and operation [1]. For critical infrastructures in e.g., energy production and transmission, transportation and public health, this transformation has led to an increased exposure to cyber, physical, and combined cyber-physical attacks. Most of these cyber-attacks have been caused by software vulnerabilities, and thus software vulnerability management has become indispensable to ensure the security of critical systems and infrastructures (e.g., safety protection systems in nuclear, high integrity control systems in transportation, etc.),

and emerging solutions with potential high impact (e.g., Artificial Intelligence, block chain, and quantum systems).

Systems are required to be more efficient whilst retaining their efficacy, resulting in a more complex security landscape. For cybersecurity, handling both hardware and software vulnerabilities throughout the system life cycle is critical [2]. To manage software vulnerabilities, Software Vulnerability Management (SVM) is a vital process to ensure the quality and security of critical systems and infrastructures.

This workshop facilitates discourse and discussions among researchers, practitioners, and students who are working on challenges and solutions related to cybersecurity, with a particular focus on sharing industry experience and project results pertaining to cyber threats on critical systems; secure software engineering; and attack detection and response mechanisms.

2. EnCyCriS/SVM Workshop

The 2024 editions of the EnCyCriS and SVM workshops were arranged as a joint workshop, bridging the industrial system focus of the EnCyCriS workshop with SVM focus on vulnerability management for software. Like previous editions of the workshops, the 4th International workshop on Engineering and Cybersecurity of Critical Systems (EnCyCriS – 2023) [3], and the 1st International Workshop on Software Vulnerability Management (SVM – 2023), the joint EnCyCriS/SVM workshop was organized in conjunction with the 46th International Conference on Software Engineering (ICSE – 2024). In the call for papers of this workshop, contributions from researchers and practitioners working on challenges and solutions for engineering and cybersecurity of critical systems and software vulnerability management were solicited. The workshop had a particular focus on sharing industry experience and project results pertaining to cyber threats on critical systems, secure software engineering, and attack detection and response mechanisms. Contributions can be in the form of position papers, research papers, and industrial experience papers.

The objectives of this workshop in terms of the Research and Development (R&D) presented in the paper contributions are to provide state of the art and practice in:

- Integrated safety and security software engineering processes and methods for CI that enable coordination among different teams and personnel with different competences, and continuous integration of new and emerging technologies enabling the 4th Industrial Revolution.
- As a means to support both development and operational phases: threat landscape for digital systems' software in critical infrastructure, and the modelling of cyber-attack scenarios for digital systems' architecture and inter-dependencies, must be considered. This includes accurate software architecture design model specification that facilitates cyber-security, Reliability, Availability, Maintainability and Safety (RAMS) assessment, SVM, and support effective communication of the threats, vulnerabilities, risks, and potential mitigation to relevant stakeholders.
- Development and use of risk models for digital systems' software for quick and effective decision-making in order to respond to fast evolving cyber incidents. For e.g., how to use, combine or tailor software system models, network topology, risk models, in a way that they could be used in incident response to determine event propagation and suitable response and mitigation strategies.
- Demonstrate the use of digital twins, Hardware-In-the-Loop (HIL) testbeds, simulators, and/or emulators developed for cybersecurity purposes in critical infrastructures, especially to understand the consequences of a cyber-attack on both the software and on the overall system-level, evaluate the effectiveness of the developed risk models, detection, and response tools and/or methods.
- Disseminate the state-of-the-art and state-of-the-practice of SVM to identify and close the gap between industry and research on the advances and practices of SVM for critical systems and infrastructures such as Artificial Intelligence based, blockchain, Augmented/Virtual/Mixed Reality, and quantum systems, as well as and the respective development paradigms, including DevOps and infrastructure-as-code.

The organization committee of this workshop included six members representing five different institutions in Norway, Korea, Australia and the UK: Institute for Energy Technology (IFE), Korea Advanced Institute of Science and Technology (KAIST), Østfold University College (HiØ), The University of Adelaide, and The University of Bristol. In this workshop, the Technical Program Committee (TPC) consisted of 26 researchers based in twelve different countries including Korea, Norway, UK, the Netherlands, Ireland, Sweden, USA, China, Canada, Australia, France, and Germany. This workshop received 19 paper submissions. Each paper was peer-reviewed by a minimum of three different reviewers, who are a part of TPC. In the end, nine papers were accepted out of 19 paper submissions. A total of 56 participants registered for the workshop.

The program for this workshop included two keynotes; the first a keynote on *"If you build it, they (probably) won't come"* by Professor Adam Joinson, University of Bath, UK. The second keynote *"Current threats and challenged for securing OT/IoT systems"* by regional operations manager Leonard Keat from Advens, France. The workshop was arranged in three main sessions, with Adam Joinson's keynote for the first session and Leonard Keat's keynote for the second and third. The workshop sessions included the nine accepted papers. The first session was on *"Developing Secure Software"*, the second session was on *"Developing Secure Software and Industrial Challenge"*, and the third session was on *"Training, Knowledge and Industrial Challenges"*. Lastly the workshop featured a panel-based discussion session arranged as a roundtable including all workshop participants. There also were questions-and-answers (Q&A) corresponding to each session mentioned above.

2.1 Keynote Talk

A keynote on *"If you build it, they (probably) won't come"* was provided by Professor Adam Joinson. Adam Joinson is full Professor at the University of Bath (UK), the director of the ESRC Digital Security by Design Social Science Hub+ ("Discribe"), and the programme lead *"risk and online behaviour"* for the national Centre for Research and Evidence on Security Threats. Joinson presented the CHERI initiative which aims to update the foundation of current insecure computing infrastructure by demonstrating that mainstream technology and software can support new security technologies based on the CHERI architecture. Its ultimate success will require overcoming HW and SW development silos, long development roadmaps, and the effort required to shift code bases to new infrastructure. This served as a backdrop for the Describe+ Hub which arranges bi-yearly sessions focusing on highlighting economic benefits of digital security by design. Here, storytelling, anthologies, and other techniques are used to imagine future possibilities and impacts of digital security by design. Adam pointed out that a challenge with security is its visibility (you don't always see it). Security doesn't really sell well. The common approach is to talk about stock market impact of security breach and Adam discussed the work Describe+ Hub is doing to address these challenges.

The second keynote, *"Current threats and challenged for securing OT/IoT systems"* was provided by Leonard Keat. Mr. Keat is a senior regional operations manager for the French company Advens. He has more than 20 years' experience in cybersecurity consultancy, training, and management, including business continuity and crisis management. His expertise includes AI security and ethics with the combination of Secure-, Privacy-, and Ethical-by-Design approach. Keats presented the current state of threats as experienced in industry, and summarized the most relevant political and geopolitical threats, as well as financially organized crime. Change in the way companies work and collaborate with third parties, as well as remote working, has led to tele maintenance becoming a common and viable attack vector. A new trend in supply chain attacks has been to gain access to a large number of sites and companies' infrastructure. Mr. Keat provided advice for how to get started with building cyber security into a company and its system and stressed the importance of 1. assessing the most important assets, functionalities, and values, 2. performing mapping and cartography to understand the infrastructure layout and functional relationships in the systems and company, and 3. prioritize and implement mitigations and improvements.

2.2 Paper Sessions

This workshop accepted 9 papers for inclusion in the workshop program and proceedings. The authors of accepted papers were invited to present their work in one of the three sessions considering the topic of their work. Session 1 was on *"Developing Secure Software"*, chaired by Awais Rashid, Session 2 was on *"Developing Secure Software and Industrial Challenge"*, chaired by Awais Rashid and John E Simensen, and the third session was on *"Training, Knowledge and Industrial Challenges"*, chaired by John E Simensen.

The following present the main key takeaways of the two papers included in Session 1.

The paper entitled *"WasmCFuzz: Structure-aware Fuzzing for Wasm Compilers"* is authored by Zhang et al. [4]. Co-author and presenter Xiaoning Du explained the central role of WebAssembly for web applications. The fact that Web Assembly Model has surpassed JIT in the number of security comments on GitHub, motivated their WasmCFuzz fuzzy testing approach. He showed an example of how to overflow a variable in the Wasm and described the approach which was able to identify 13 previously unidentified bugs in mainstream browsers within just a week.

The paper entitled *"Mitigating Security Issues in GitHub Actions"* is authored by Delicheh and Mens [5]. Delicheh explained how collaborative practices have revolutionized the software development

process. He showed how to commit and run actions on projects on GitHub, as well as describing how workflows works. A challenge with GitHub is that reusable components again can use other components. Delickeh gave examples of security risks associated with misconfigurations of repositories and workflows, before suggesting several practical mitigations, including dependency and security monitoring, pinning actions, strict access control and verified creator practice to name some. It was concluded that there is a need to integrate security seamlessly into the automated collaborative software development processes.

In Session 2, two papers were presented. These papers are summarized in the next paragraphs.

The paper entitled *"Trust in Software Supply Chains: Blockchain-Enabled SBOM and the AIBOM Future"* is authored by Xia et al. [6]. Xia explained that the recent spike in supply chain attacks, in conjunction with a general lack of transparency into the software supply chain, are motivator for work that is done on Software Bills Of Material (SBOM). To increase trust in the SBOM, a block-chain based approach to keep track of SBOMs was suggested. A focus of the discussions following the presentations was on industry willingness to share information, and on points in the process where human actions and decisions are needed.

The paper entitled *"Interplay of Digital Twins and Cyber Deception: Unraveling Paths for Technological Advancements"* is authored by Heluany et al. [7]. Co-author Amro gave the results of a literature survey and market analysis, reporting a lack of mature scientific and commercial contributions in Digital Twin and cyber deception technologies. Examples of gains from digital twin-based honeypots, such as cyber deception, anomaly detection, and threat intelligence, was provided. Two approaches were discussed; using digital twins as honeypots or using the data itself as a honeypot and feed it to a digital twin to discover effects. One suggestion from the work was to develop and train a large language model as attacker response scheme.

In Session 3, five papers were presented. These papers are summarized in the next paragraphs.

The paper entitled *"Building a Cybersecurity Knowledge Graph with CyberGraph"* is authored by Falcarin and Dainese [8]. Falcarin explained the challenge that cyber security information sources, e.g., information on known vulnerabilities, newly identified weaknesses, and threats, mitigations, etc., are spread across different places and repositories. To support developers in this effort, the CyberGraph tool was presented. CyberGraph automatically builds and updates a single, easily queryable cybersecurity knowledge graph. It automatically links heterogeneous data from different public repositories. A discussion item was to which extent the tool can aid stakeholders in prioritizing between CVEs to fix, and the order in which you should apply fixes according to severity of the CVE or potential impact on the system.

The paper entitled *"Training Developers to Code Securely: Theory and Practice"* is authored by Ryan et al. [9]. Ryan explained the importance of software design and coding practices with regards to (avoiding) producing vulnerabilities. Positive features of effective secure development training were identified through a literature analysis of 109 papers. These were compared to training information from a large developer survey (n=962) to assess how training in the field matches these key positive features. Ryan explained that only 39% had been offered cyber security coding training. Of these only 60% reported the training was relevant. She concluded that although a few developers experience excellent secure-coding training, some receive inadequate training, the majority receive no training at all.

The paper entitled *"On DevSecOps and Risk Management in Critical Infrastructures: Practitioners' Insights on Needs and Goals"* is authored by Ramaj et al. [10]. Ramaj gave a short introduction to DevSecOps and explained the approach with regards to bolstering security of software applications by continuously changing an organization's approaches to

tackle emerging threats. She presented the results of a qualitative study to identify needs and goals of implementing DevSecOps in critical infrastructures from the perspective of DevOps, developers, and security experts. The authors had selected a Grounded Theory approach (Glaser) for data analysis. It was reported that most of the problems experienced arise from the interdependencies systems have with each other, and that the sheer number of systems and system variations is challenging. Discussion after the presentations was related to the prioritization of the needs reported from the participants in the study - e.g., choose and rank the five most pressing needs and challenges.

The paper entitled *"Cyber-incident Response in Industrial Control Systems: Practices and Challenges in the Petroleum Industry"* is authored by Gnanasekaran et al. [11]. Gnanasekaran presented findings from ongoing research into cyber-incident response as seen from a more holistic view. She explained the need to take a holistic approach of cyber-incident response in industrial control systems and investigate relevant stakeholders such as e.g., internal roles, and external actors, such as Security Operations Centers and Computer Security Incident Response Teams. A total of seven semi-structured interviews were performed with petroleum industry to identify roles, practices, and challenges in cyber-incident response. The result of the interviews was a mapping and description of actors' roles, tasks and placement presented as a stakeholder diagram.

The paper entitled *"Cybersecurity and medical devices: a bull in a china shop"* is authored by Filippini and Spiller [12]. Filippini presented experiences from the medical industry with regards to analyzing, managing, and maintaining cybersecurity of a complex medical device along its lifecycle. He introduced a critical medical device and explained that until 2017, medical devices were exempt from being secure (not mandated). The current view now however is that it is expected that the medical devices themselves will be specific targets for cyber, which is reflected in Medical Device Regulation (MDR) 2017/745 which states that the manufacturer is responsible for cybersecurity. Filippini gave examples of how they work to integrate safety and security within medical devices, and he shared challenges as seen from a manufacturer.

3. DISCUSSION

This workshop proposed one discussion slot that lasted for one hour at the end of the last session. Authors and presenters were invited to answer the questions from the session chair and discuss more freely around the thematic of the workshop sessions.

The keynotes highlighted several challenges regarding cyber security for development and operation of software digital systems and critical infrastructure. During the discussion session, the industrial challenges, represented by one keynote and several presenters, was given a central focus. Through discussion it was revealed that state of practice in industry does not always correspond to the academic view of where industry stands. What can be considered straightforward in principle, e.g., asset identification and criticality evaluation, can be a struggle for industries for several reasons including time constraints, lack of personnel and competence, lack of prioritization etc. At the same time, academia suggests approaches and methods that could benefit industry if developed further and aligned with more concrete industrial needs. One example is the use of Artificial Intelligence (AI) such as Machine Learning and Deep Learning models for various Software Vulnerability Management tasks that has recently gained traction in the academic literature. However, the adoption and efficacy of these AI solutions in industrial environments is still limited, often due to lack of practical evaluations. Industry is often faced with proving the cyber-security adequacy of the systems and organizations they have with regards to regulations and standards. They are often stuck with their existing systems and solutions, and creating ideal solutions from scratch is not an option, instead making smaller incremental changes and improvements is often only viable option. Academia seems to have a more idealistic view and often suggest solutions that would not be realistic from a cost, resource, or competence point of view for industry.

Secure coding was discussed in more detail, including how to train developers to code securely, how to create awareness through using frameworks, and challenges regarding the use of AI for secure coding to name some. From the industry side, a discussed challenge was that some regulators seem to be more focused on being compliant with laws and regulation, seemingly as a paper exercise, rather than discussing and dealing with deviations and correcting measures in a more detailed approach.

The raised challenges ask for more collaborations between academia and industry, for example through observational studies, to identify and incorporate real-world requirements while designing, developing, and evaluating solutions for engineering and securing critical infrastructures and systems.

4. CONCLUSIONS

Like the first four editions of EnCyCriS, and the first edition of SVM, the joint EnCyCriS/SVM workshop was organized in conjunction with ICSE. This workshop accepted nine papers for presentation during the workshop in addition to two keynote speakers from academia and industry. The first keynote *"If you build it, they (probably) won't come"* highlighted the challenge of making security more visible and how to work towards empowering industry and society at large to imagine future possibilities and impacts of digital security. The second keynote *"Current threats and challenges for securing OT/IoT systems"* discussed the current state of threats as experienced in industry, and summarized the most relevant political and geopolitical threats, as well as financially organized crime. The keynotes served as backdrop to facilitate more specific discussion between academics and practitioners on this area.

The accepted papers covered different topics including:

- Safe, reliable, and secure by design - and - Safety and security co-engineering.
- Software Vulnerability Management for critical and emerging systems, including threat modeling and software vulnerability detection and analysis.
- Cyber response estimation on software and hardware of CI using models, simulations, and digital twins.
- The role and impact of human in cybersecurity in critical infrastructures development and operation.
- Human factors in cybersecurity software engineering and software vulnerability management.

A sixth edition EnCyCriS workshop and a third edition of the SVM workshop are planned to be arranged in conjunction with ICSE 2025 – Ottawa, Canada, in the period April 27th-May 3rd, 2025.

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7th International Workshop on Software-intensive Business (IWSiB 2024): Software-intensive Business in the Era of Generative Artificial Intelligence

Jorge Melegati

Free University of Bozen-Bolzano
Bolzano, Italy

jorge.melegati@unibz.it

Dimitri Petrik

University of Stuttgart
Stuttgart, Germany

dimitri.petrik@bwi.uni-stuttgart.de

Andrey Saltan

LUT University
Lahti, Finland

andrey.saltan@lut.fi

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ABSTRACT

The 7th International Workshop on Software-intensive Business (IWSiB'24), held on April 16, 2024, in Lisbon, Portugal, as part of the 46th International Conference on Software Engineering (ICSE 2024), brought together diverse research communities to address the pressing challenges faced by software-producing organizations. These challenges, driven by changing demands, evolving technologies, and dynamic environments, require interdisciplinary collaboration across software engineering and business research. The workshop focused on the theme “Software Business in the Era of Generative Artificial Intelligence,” highlighting the transformative impact of GenAI and large language models on software businesses. The event featured a keynote by Jan Bosch and showcased 11 accepted papers on a diverse range of topics. Participants engaged in interactive discussions, group activities, and ideation sessions aimed at identifying future challenges and solutions for software-intensive businesses, fostering ongoing community development and cross-disciplinary synergy.

Categories and Subject Descriptors

•Social and professional topics~Professional topics~Management of computing and information systems •Social and professional topics~Professional topics~Computing and business •Software and its engineering~Software creation and management •Software and its engineering~Software creation and management~Collaboration in software development •Software and its engineering~Software creation and management~Software development process management~Software development methods~Agile software development •Software and its engineering~Software creation and management~Software verification and validation~Software defect analysis~Software testing and debugging

General Terms

Management, Economics, Experimentation, Security, Human Factors.

Keywords

Software-intensive business, artificial intelligence, agile, software ecosystem, ethics.

1. INTRODUCTION

Software-producing organizations are increasingly challenged by rapidly changing demands, evolving technologies, and dynamic ecosystems in which their products and services must operate [1]. To succeed, these organizations must adapt their operating models and leverage emerging trends [6]. Approaches like design thinking [4], lean startup [13], agile development, and DevOps [5] enable rapid identification of business value and swift adaptation to changes. The advent of large language models (LLMs), such as ChatGPT, represents a significant shift, with the potential to deeply transform software

development [3] and disrupt software-intensive businesses by introducing greater flexibility in experimentation and decision-making.

The challenges to making software organizations successful and sustainable are multi-disciplinary [1]. Technical challenges in software engineering, such as managing requirements, platforms, and complex technology landscapes, must be balanced with the need for user adoption and effective business models. These challenges span various sectors, affecting organizations from startups to industry giants. The field of software-intensive business research addresses these issues by exploring software-based value creation and delivery, both within and between organizations, complementing software engineering with a business-driven perspective.

The 7th International Workshop on Software-intensive Business (IWSiB'24), hosted as part of the 46th International Conference on Software Engineering (ICSE 2024), convened researchers, practitioners, and industry leaders to explore the future of software-intensive business, with a particular focus on AI, digitalization, and continuous value delivery. The workshop commenced with a keynote by Jan Bosch, providing valuable insights into the strategies and technologies shaping the industry. From 20 submissions, 11 papers were accepted after a rigorous review process, showcasing diverse and innovative research on the impact of generative AI and large language models on software engineering and business practices. As businesses continue to navigate the complexities of digital transformation, the discussions and frameworks presented at this workshop will play a crucial role in guiding successful and sustainable innovations in software-intensive domains.

2. KEYNOTE

Jan Bosch, Professor at the Department of Computer Science and Engineering, Chalmers/University of Gothenburg and Director of the Software Center¹, an organization that is the collaboration center between more than a dozen software-intensive companies and five Swedish universities delivered an insightful keynote on the imperative need for digitalization in modern businesses.

In his keynote, Bosch highlighted the stark reality of business longevity in the face of digital disruption, referencing that many companies on the Fortune 500 list from previous decades have not endured. He underscored that failure to innovate and adapt to new digital paradigms, particularly those involving software, data, and AI, is a primary reason for this decline. Bosch emphasized the concept that “disruption should be the new normal,” suggesting that companies must continuously innovate and develop new core capabilities to stay competitive.

¹ <https://www.software-center.se>

Bosch further delved into the specifics of digitalization, focusing on the need for businesses to organize around value and transition from requirement-driven to outcome-driven approaches. He introduced models and frameworks like the Stairway to Heaven, CIVIT, CAFFEA, and HYPEX, which help companies integrate and accelerate digital practices. He also discussed the benefits of shorter value delivery cycles and continuous deployment models. By citing examples from companies within the Software Center and their successful adoption of these practices, Bosch illustrated how a holistic approach encompassing technical, organizational, and business aspects can significantly enhance a company's digital transformation and ensure sustained value delivery to customers.

3. THE WORKSHOP

IWSiB'24 was organized into three distinct sessions, each thematically structured around the accepted papers. These sessions explored three critical areas of research within the software-business domain: AI & Software-intensive business, Agile & software-intensive business, and Software ecosystems. Each session provided valuable insights into the evolving landscape of software-intensive businesses, highlighting the latest advancements and challenges in these fields.

3.1 AI & Software-Intensive Business

This session consisted of four presentations and focused on the integration and impact of artificial intelligence within software-intensive businesses, exploring how AI can drive innovation and efficiency across various domains.

1. The paper “Artificial Intelligence in the Public Sector – An Agenda for Responsible Innovation through Learning” by Leif Z. Knutsen, Jo E. Hannay and Michael A. Riegler outlined a comprehensive agenda for the responsible implementation of AI in public services [11]. The authors propose an adaptive learning framework to address the rapidly evolving technology landscape, emphasizing cross-sectional solutions and institutional learning across functional, organizational, geographic, and national boundaries.
2. The study “Human-AI Collaboration in Software Engineering: Lessons Learned from a Hands-On Workshop” by Muhammad Hamza, Dominik Siemon, Muhammad Azeem Akbar and Tahsinur Rahman investigated the dynamics of human-AI collaboration using ChatGPT [7]. Through a workshop involving professional software engineers, the authors explore the evolving nature of human-AI interaction, the capabilities of AI in software engineering tasks, and the challenges and limitations of integrating AI into development processes.
3. The paper, “Towards Stability, Predictability, and Quality of Intelligent Automation Services: ECIT Product Journey from On-premise to As-a-Service” by Damian Kedziora, Dominik Siemon, Edona Elshan and Monika Soñta explored the transformation of ECIT's intelligent automation services from traditional on-premise models to a modern SaaS approach [10]. The authors discuss the triggers, journey, and implications of this transition on cost, quality, and customer satisfaction, highlighting the benefits of the ‘Robot-as-a-Service’ model.
4. Finally, the paper “Researchers’ Concerns on Artificial Intelligence Ethics: Results from a Scenario-Based Survey” by Marianna Jantunen, Richard Meyes, Veronika Kurchyna,

Tobias Meisen, Pekka Abrahamsson and Rahul Mohanani presented a survey with 33 researchers working with AI regarding negative aspects of this technology [9]. Based on five scenarios depicting fictional AI systems, the analysis of the answers led to several potential threats divided into three categories: social & ethical, technical & design, and safety & security.

3.2 Agile and Software-Intensive Business

This session delved into the challenges and strategies of implementing agile methodologies within software-intensive environments, emphasizing the balance between agility and other critical business factors like security and testing.

1. The paper “How To Get Good At Data: 5 Steps” by Helena Holmström Olsson and Jan Bosch presented a maturity model for the adoption of data-driven practices divided into five steps [8]. The model was derived from a multiple-case study on 11 companies. The authors also describe how companies could use these steps to guide their transition to fully data-driven practices.
2. The paper “Developing a Taxonomy for Agile Scaling Frameworks” by Yagmur Turhan, Deborah Buehrle and Georg Herzwurm presents a taxonomy for scaling agile frameworks based on the dimensions of dissemination, applicability, implementation, components, processes, and continuous improvement [15]. For the evaluation of the taxonomy, the authors employ it to compare Scrum@Scale and LeSS.
3. The paper “Unveiling Customer Needs: A Comprehensive Exploration of Jobs to be Done Interviews” by Esra Özal and Jürgen Münch investigates the Jobs to be Done (JTBD) framework as a method for understanding customer needs in software product management [12]. Through qualitative interviews, the study identifies critical elements and challenges in conducting JTBD interviews, such as thorough preparation, skillful communication, trust building, and structured interview approaches. The findings provide practical guidance for companies aiming to align their product development with real customer needs, thereby enhancing innovation and customer-centric product development practices.
4. The study “Security Practices in Agile Software Development: A Mapping Study” by Alejandra Selva-Mora, and Christian Quesada-López explored the integration of security practices within agile software development environments [14]. By mapping 36 primary studies, the authors identify 255 security activities categorized into the Building Security in Maturity Model (BSIMM). The study highlights the benefits, such as increased security awareness and implementation, and the challenges, like the lack of knowledge and complexity in aligning security with agility. The findings underscore the need for empirical evaluation of these practices in real-world settings and provide a foundation for future research on security in agile methodologies.

3.3 Software Ecosystems

This session examined the complexities of managing software ecosystems, discussing the tools and techniques for optimizing ecosystem health, fostering innovation, and sustaining growth in competitive markets.

1. The paper “Modeling 6G Software Business Ecosystem: A Look Ahead” by Nan Yang, Dominik Siemon, and Sami Hyrynsalmi described a study employing methodologies from the future studies field to analyze the 6G ecosystem from the software-intensive business perspective [17]. The authors identified four building blocks, namely vision, stakeholders, their complex relationships, and key opportunities and challenges.
2. The paper “Software Ecosystem Orchestration with Topic Modeling” by Casper van Schothorst, Robbert Schuurmans, and Slinger Jansen presented an approach for orchestrating software ecosystems using topic modeling to analyze extensive application data [16]. The method supports ecosystem orchestrators by providing strategic insights into the ecosystem’s health, identifying new domains for platform growth, and discovering potential partners. The authors apply this approach to Slack and Microsoft Teams, comparing their app stores to illustrate competitive analysis, app classification, and ecosystem evolution. The results demonstrate how topic modeling can effectively inform strategic decisions, optimize app portfolio management, and support continuous ecosystem growth and health monitoring.
3. The study “Bridging the Gap: Addressing Software Testing Challenges in Namibian Startups through a Tailored Training Approach” by Hilma Aludhilu and Erkki Sutinen identified key software testing challenges faced by Namibian startups, such as unfamiliarity with testing methods, insufficient use of testing tools, and lack of test coverage [2]. To address these issues, the authors propose a tailored training program focusing on awareness, skills, and attitude, structured around Bloom’s Taxonomy framework. The training approach combines theoretical knowledge and empirical data to enhance developers’ testing capabilities, aiming to improve product quality and support the sustainable growth of Namibian software startups.

4. CONCLUSION

IWSiB’24 underscored the critical importance of AI, digitalization, agile methodologies, and ecosystem management in shaping the future of software-intensive businesses. The workshop successfully provided a platform for the software engineering and business research communities to discuss the unique challenges these businesses face, share early-stage work, and gain valuable feedback from a diverse audience. By facilitating exchanges between academics and practitioners, the workshop highlighted best practices, solutions, and contemporary challenges, while also working to build a joint research agenda that addresses the intricate interplay between software engineering and business.

The keynote by Jan Bosch set a strategic tone, emphasizing the need for continuous innovation and alignment within organizations. The presented papers contributed innovative solutions and empirical insights into key challenges, such as integrating security into agile practices and optimizing software ecosystems. These contributions are crucial as businesses navigate digital transformation and the rise of Generative Artificial Intelligence, helping ensure sustained growth and competitive advantage in the software industry.

The IWSiB series has played a pivotal role in uniting various research communities under the umbrella of software-intensive business. It has become a leading forum for nurturing emerging researchers and integrating them into the broader software engineering community. The workshop continues to be a vital avenue for advancing research and practice in this growing field, driving the development of new

approaches and solutions that address the evolving needs of the software industry.

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First International Workshop on Quantum Software Engineering: The Next Evolution (QSE-NE) Summary 2024

Arif Ali Khan

M3S Empirical Software Engineering Research Unit, University of Oulu
Oulu 90014, Finland
arif.khan@oulu.fi

Muhammad Azeem Akbar

Software Engineering Department, LUT University
Lahti 15100, Finland
azeem.akbar@lut.fi

Peng Liang

School of Computer Science, Wuhan University
Wuhan 430072, China
liangp@whu.edu.cn

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As we enter the new quantum age, tech giants like IBM, Google, and Microsoft are leading the way. But, there is a major challenge: we need to turn exciting quantum possibilities into real, practical tools. That's where Quantum Software Engineering (QSE) comes in, and it's why "Quantum Software Engineering: The next Evolution (QSE-NE)" workshop is proposed. This workshop is conducted to help guide this change. It aims to highlight the latest ideas in QSE and get people from different fields talking, from university researchers to big tech companies. Our main goals were to share new QSE methods, tools, and studies and to tackle the tough problems we face in developing quantum software. This workshop is all about setting the direction for the future of building quantum software, at a time when it is really crucial. More details can be found at: <https://sites.google.com/view/qse-ne/home>.

1. INTRODUCTION

The quantum-based industrial revolution is already looming, and the twenty-first century will probably be the "quantum age". The field of Quantum Computing (QC) has received rapidly growing industrial interest to an even greater extent. It is evidenced by the fact that technology giants such as IBM [6], Google [5], and Microsoft [8] have heavily invested in and committed to QC. It is well recognized that disruptive solutions empowered with QC have opened the way for business development in various areas, such as cybersecurity, financial services, and aerospace. However, cost-effectively, intuitively, and efficiently developing highly dependable QC applications is exceptionally challenging and remains an unsolved problem [7, 11]. However, enabling QC to live up to its promising potential - only quantum mechanics, mathematics, hardware, and quantum computer science are not enough, a new genre of "Quantum Software Engineering" (QSE) is becoming a must. A national agenda for software engineering research and development presented by SEI - Carnegie Mellon University explicitly considered "Engineering Quantum Computing Software Systems" as a future focus area - "Advances in software engineering for quantum are as important as the hardware advances. The goals of this research area are to first enable current quantum computers to be programmed more easily and reliably, and then enable increasing abstraction as larger, fully fault-tolerant quantum computing systems become available. A key challenge is to eventually, fully integrate these types of systems into a unified classical and quantum software development life cycle" [2].

To perceive this, we proposed the workshop to provide a venue to discuss novel Quantum-specific software engineering tools, processes, techniques, frameworks, models, and best practices. This

workshop will bring together researchers and practitioners from academia and industry to present research work to ensure the quality of QSE life cycle activities and improve overall productivity.

This workshop welcomes submissions from industrial experts and academic researchers on quantum software engineering. The scope is not limited to specific QSE practices but includes the broad level research studies that encapsulate the overall QSE life cycle defining novel processes, tools, models, standards, practices, and frameworks to maximize the significance of quantum software systems.

2. REVIEW PROCESS AND PROGRAM

The review process for the workshop was designed to ensure the highest quality of contributions and fairness. To achieve this, we employed a double-anonymous review process, where both external reviewers and authors remained anonymous to each other. Each submission was reviewed by at least three experts in the field, and papers submitted were required to omit any information that could reveal the authors identities. This rigorous and impartial process ensured a balanced and unbiased evaluation of all submissions.

The QSE-NE 2024 is the first addition of this workshop series, co-located with the ACM International Conference on the Foundations of Software Engineering (FSE) 2024, Mon 15 - Fri 19 July 2024 Porto de Galinhas, Brazil. It was a half day workshop, consisting of two sessions and took place on the 15th of July, started at 09:00am and finished at 12:30PM. The details schedule of the workshop program is provided at: <https://2024.esec-fse.org/program/program-fse-2024/>.

We received a total of ten submissions from different countries, after thoroughly reviewing each submission by the program committee members of which six were accepted for presentation, resulting in a 60% acceptance rate. For every accepted paper, it was mandatory for at least one of the authors to orally present their work. All accepted papers are included in the ACM digital library.

3. ACCEPTED PAPERS

The accepted papers were presented in two different sessions. The morning session began with an opening note from the Chair, Muhammad Azeem Akbar, who highlighted the significance of the workshop and the overall process of conducting it. Following the opening note, the article - *QCSHQD: Quantum Computing as a Service for Hybrid Classical-Quantum Software Development: A Vision* [10] was presented. In this article, the authors proposed the idea of hybrid (quantum-classical) computing by pro-

viding quantum computers as a service to classical developers. The proposed framework is designed based on the concepts of service-oriented computing, allowing classical developers to leverage the potential benefits of quantum computing without needing in-depth knowledge of it. The vision presented in this article provides a roadmap for groundbreaking innovations by addressing the key challenges of hybridizing classical and quantum computers.

In the second presentation, the article - *Classiq: Towards a Translation Framework to Bridge the Classical-Quantum Programming Gap* [3] is introduced. This article proposes a vision for the *Classiq* translation framework, which aims to bridge the gap between classical and quantum computing. The core idea behind the proposed framework is to translate high-level classical programming code (Python) into low-level quantum code, such as quantum assembly. This framework provides a platform for classical programmers who lack prior quantum experience. The proposed framework offers an interface that allows classical developers to write classical code, which is then translated into quantum code. This translated code is executed on available quantum computers (as a service) using the API gateway, and the results are fetched and translated back into a format understandable by classical computers. These translated results are delivered to the classical developers in the final phase.

In the third presentation of the morning session, Daniel et al. [4] presented the article - *Gate Branch Coverage: A Metric for Quantum Software Testing* - where a new testing metric called Gate Branch Coverage is introduced. This metric is designed to provide insights into the verification status of quantum programs and enhance the overall quantum testing process. Gate Branch Coverage specifically examines the properties of quantum controlled-type gates, measuring the number of branches exercised during the execution of quantum programs.

In the last paper of the morning session - *A Conceptual Framework for quantum Integration Challenges in 6G Technology* [9], the authors investigated the challenges of integrating quantum computing into 6G technology. It highlights a significant impact of quantum computing on the future of telecommunications and identifies key application areas within 6G through a thorough analysis of existing literature. Ultimately, it underscores the transformative potential of QC within 6G and sets the stage for further research in this rapidly evolving field.

The first session concluded with a coffee break, followed by the second session of paper presentations from 11:00 AM to 12:30 PM, during which two articles were presented. In the first article of the session - *Integrating quantum Computing and Blockchain: Building the Foundations of Secure, Efficient 6G Technology* [12], a framework is proposed to integrate quantum computing and blockchain to enhance city management for the 6G era. It leverages quantum computing's processing power and blockchain's security to take advantage of 6G's high speed, low latency, and massive connectivity. The study uses a mixed-methods approach, including expert interviews, literature reviews, and case studies, to develop a strategic framework addressing integration challenges and evaluating its impact on city management. The research highlights critical issues like quantum computing complexity and blockchain scalability, aiming to make city management smarter, faster, and more secure.

In the last paper - *Quantum Software Engineering: A New Genre of Computing* [1], the authors discussed the transition of quantum computing from theoretical science to a commercially viable technology. It highlights the need for software-intensive methodologies and tools to develop quantum applications as QC hardware advances. The paper outlines the life cycle of quantum software engineering (QSE), including requirements engineering, design, implementation, testing, and maintenance.

4. CONCLUSIONS AND FUTURE THOUGHTS

The QSE-NE workshop has successfully established a significant platform for discussing the advancements and challenges in the field of Quantum Software Engineering (QSE). By bringing together experts from academia and industry, the workshop facilitated the exchange of innovative ideas, methodologies, and frameworks that are crucial for the development of robust quantum software. The high quality of accepted papers, which addressed various aspects of QSE from hybrid classical-quantum computing to quantum software testing metrics, underscored the diverse and interdisciplinary nature of this emerging field.

Looking ahead, future iterations of this workshop will aim to broaden the scope and depth of the discussions by including more hands-on sessions, collaborative projects, and industry-academic partnership initiatives. These will focus on practical implementation strategies, real-world applications, and the integration of quantum computing with other advanced technologies like blockchain and 6G. Additionally, the workshop will expand its focus to include more comprehensive training and tutorials for participants, fostering a deeper understanding of quantum computing concepts and their practical applications. This will ensure that QSE-NE continues to be at the forefront of guiding the evolution of quantum software engineering.

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SEA4DQ 2024 Workshop Summary

Tim Menzies, Bowen Xu
(General Co-Chair)
North Carolina State University, USA
{tjmenzie,bxu22}@ncsu.edu

Hong Jin Kang
(Program Co-Chair)
University of California, Los Angeles, USA
hjkang@cs.ucla.edu

Jie M. Zhang
(Program Co-Chair)
King's College London, UK
jie.zhang@kcl.ac.uk

Jiri Gesi
(Industrial Co-Chair)
Amazon Science, USA
{auth3,auth4}@auth34.org

Sagar Sen
(Industrial Co-Chair)
SINTEF, Norway
Sagar.Sen@sintef.no

Beatriz Cassoli
(Industrial Co-Chair)
TU Darmstadt, Germany
beatrizcassoli@gmail.com

Nicolas Jourdan
(Web Co-Chair)
TU Darmstadt, Germany
N.Jourdan@ptw.tu-darmstadt.de

Jieke Shi
(Web Co-Chair)
Singapore Management University, Singapore
jiekeshi@smu.edu.sg

Phu Nguyen
(Publicity Co-Chair)
SINTEF, Norway
Phu.Nguyen@sintef.no

Valentina Golendukhina
(Publicity Co-Chair)
University of Innsbruck, Austria
Valentina.Golendukhina@uibk.ac.at

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Welcome to the sixth edition of the workshop on Machine Learning Techniques for Software Quality Evaluation (SEA4DQ 2024), held in Brazil, July 16th, 2024, co-located with ESEC / FSE 2024 [1]. Three papers from all over the world were submitted, and all of them were accepted based on the overall positive reviews. The program also featured two keynotes by Denys Poshyvanyk on *Towards an Interpretable Science of Deep Learning for Software Engineering: A Causal Inference View* and Qinghua Lu on *Responsible AI Engineering from A Data Perspective*.

1. INTRODUCTION

Modern systems are centered on data, using data in novel and intelligent ways. One key driver for increased data availability is the availability of data-sharing platforms and open data portals, which are gaining momentum. Another key driver is the Internet of Things (IoT) being realized in many industrial sectors and application domains facilitating data acquisition from physical sensors and devices on an unprecedented scale.

Data quality is crucial, as the data acquired and used by modern software systems strongly impacts on the reliability, robustness, efficiency, and trustworthiness of these systems. As an example, the quality of the data acquired and used for decision support is a common factor across industrial sectors that underpin our reliability and trust in IoT and thus cyber-physical systems (CPS).

Data quality can deteriorate due to several factors. For example, IoT/CPS systems can experience sensor faults and failures due to operating in harsh and uncertain environments, bias in measurement in digital health, intermittent loss in connectivity due to physical barriers or unreliable communication protocols, and inconsistencies in data being transformed/pre-processed and duplicated. Poor data quality reduces our trust and reliance on such systems.

Software engineering techniques need to take data quality management seriously as software systems are increasingly becoming

data-centric. These software engineering techniques need to interact hand in hand with AI models to pre-process data, detect anomalies in data quality within both short-term streaming data and long-term historical data and to repair erroneous data, replace missing data, and detect ethical issues such as bias in data.

The workshop aims to provide a forum for researchers and practitioners to exchange and discuss trending views, ideas, state-of-the-art, work in progress, and scientific results highlighting aspects of software engineering and AI to address the problem of data quality. Topics include but are not limited to:

- Requirements engineering for data quality;
- Software architectures and frameworks for data quality management;
- Software and AI solutions for data ingestion, collection, and acquisition for the sake of data quality;
- Software and AI solutions for Data pre-processing; filtering, labelling, cleaning for the sake of data quality;
- Software tools for data quality management, testing, and profiling;
- Quantification of data quality;
- Case studies evaluating existing data quality technique on real systems;
- Trade-off between data quality, performance, and accuracy.

2. KEYNOTE TAKEAWAY

Denys Poshyvanyk is a Chancellor Professor and a Graduate Director in the Computer Science Department at William & Mary, United States, delivered the keynote entitled “*Towards an Interpretable Science of Deep Learning for Software Engineering: A Causal Inference View*” [2]. The talk mainly introduces DOCODE, a post hoc interpretability framework specific to NCMs that can explain model predictions. DOCODE is based upon causal inference to enable programming language-oriented explanations. While the theoretical underpinnings of DOCODE are extensible to exploring different model properties, Denys provided a concrete

instantiation that aims to mitigate the impact of spurious correlations by grounding explanations of model behavior in properties of programming languages. doCODE can generate causal explanations based on Abstract Syntax Tree information and software engineering-based interventions. To demonstrate the practical benefit of doCODE, Denys also presented empirical results of using doCODE for detecting confounding bias in NCMs.

Qinghua Lu is a principal research scientist and leads the Responsible AI science team at CSIRO's Data61, Australia, delivered the keynote entitled "*Responsible AI Engineering from A Data Perspective*" [3]. The talk illustrates a responsible AI engineering approach to address system-level responsible AI challenges. This includes engineering/governance methods, practices, tools, and platforms to ensure responsible AI and AI safety. Specially, Qinghua also shared her extensive experience on how the responsible AI engineering approach tackles data challenges within the context of responsible AI.

3. INSIGHTS FROM PAPER SESSIONS

Akond from Auburn University presented his work on evaluating the quality of open source Ansible Playbooks [4]. He presented an empirical study with 2,952 OSS Ansible playbooks where he evaluated the quality of OSS playbooks from the perspective of executability, i.e., if publicly available OSS Ansible playbooks can be executed without failures. From the empirical study, he observed 71.5% of the mined 2,952 Ansible playbooks cannot be executed as is because of four categories of failures.

Liming from Data61, CSIRO presented her work on a pilot study in surveying data challenges of automatic software engineering tasks [5]. Her survey emphasizes the pivotal role of data in SE research. To aid understanding, she summarized the actual data challenges encountered by SE tasks and demonstrate how LLMs can effectively address these challenges. These examples offer evidence of LLMs' enhanced performance and generalization capabilities. Despite the continuous emergence of numerous new LLM solutions, LLM application in SE remains in its early stages. Although LLMs have made significant strides in SE compared to traditional methods, new data challenges persist. Future work will need to continually explore and investigate new data issues in LLM for SE research.

Yi from Nanyang Technological University presented his work on jailbreaking ChatGPT via prompt engineering [6]. The work mainly addressed a research question *With the evolution of LLMs, how does the effectiveness of jailbreak prompts change?* To answer the question, Yi embarked on an empirical study targeting the LLMs underpinning ChatGPT, one of today's most advanced chatbots. His methodology involved categorizing 78 jailbreak prompts into 10 distinct patterns, further organized into three jailbreak strategy types, and examining their distribution. He assessed the effectiveness of these prompts on GPT-3.5 and GPT-4, using a set of 3,120 questions across 8 scenarios deemed prohibited by OpenAI. Additionally, his study tracked the performance of these prompts over a 3-month period, observing the evolutionary response of ChatGPT to such inputs. His findings offer a comprehensive view of jailbreak prompts, elucidating their taxonomy, effectiveness, and temporal dynamics.

4. CONCLUSIONS

There is an ongoing trend of establishing high-quality data for AI system from software engineering perspective. This observation is manifested by the continually growing number of studies published in journals and conferences and the diversity of approaches

and techniques to model and evaluate data quality. As part of a selection of workshops addressing associated topics, SEA4DQ further indicates the need to coordinate the efforts in provisioning a lasting forum for exchanging ideas and presenting the results.

Following past editions, to help increase the visibility of the workshop and further bring together researchers in this area, the 4th edition of the SEA4DQ workshop will also be extended through a special issue of the Empirical Software Engineering journal. We invite extended and revised versions of the papers presented at the workshop, as well as newly submitted, original work.

Finally, as organizers of SEA4DQ 2024, we would like to warmly thank all authors for submitting their quality work, all reviewers for providing insightful comments, and all participants for their contribution to the dynamics of the workshop. We also thank the SEA4DQ steering committee for entrusting us with the organization of this edition.

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The Trailer of the ACM 2030 Roadmap for Software Engineering

Mauro Pezzè

USI Università della Svizzera Italiana, Lugano, Switzerland
Constructor Institute of Technology, Schaffhausen, Switzerland
Università degli studi di Milano Bicocca, Milano, Italy
mauro.pezze@usi.ch

Luca Di Grazia

USI Università della Svizzera Italiana
Lugano, Switzerland
luca.di.grazia@usi.ch

Matteo Ciniselli

USI Università della Svizzera Italiana
Lugano, Switzerland
matteo.ciniselli@usi.ch

Niccolò Puccinelli

USI Università della Svizzera Italiana
Lugano, Switzerland
niccolo.puccinelli@usi.ch

Ketai Qiu

USI Università della Svizzera Italiana
Lugano, Switzerland
ketai.qiu@usi.ch

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The landscape of software engineering has dramatically changed. The recent advances in AI, the new opportunities of quantum computing, and the new challenges of sustainability and cyber security upset the software engineering research prospective. The 2030 SE-Roadmap special issue of ACM TOSEM Transactions on Software Engineering and Methodology gives a 360° view of the research challenges of the 30ties with a thorough editorial, four roadmap papers from the ACM TOSEM editorial board, and over 30 peer-reviewed papers from the research community.

This paper previews the main content of the 2030 roadmap special issue with a report from the 2030 SOFTWARE ENGINEERING ROADMAP workshop, co-located with ACM SIGSOFT FSE Foundations of Software engineering on July 15th and 16th, 2024 in Porto de Galinhas, Brazil, that spotlighted the software engineering research horizon to feed ideas into the ACM TOSEM special issue. The paper discusses the new challenges to software engineering that emerged in the SE2030 workshop: AI for software engineering, software engineering by and for humans, sustainable software engineering, automatic programming, cyber security, validation and verification, and quantum software engineering.

1. INTRODUCTION

The landscape of software engineering has dramatically changed, and we need a new roadmap for the research in software engineering. In this paper we overview the results of the 2030 SOFTWARE ENGINEERING ROADMAP workshop that was co-located with ACM SIGSOFT FSE Foundations of Software engineering on July 15th and 16th, 2024 in Porto de Galinhas, Brazil. The liberating structure¹ format of the intensive two-full-days discussion among the over sixty participants offers an unbiased viewpoint of the main challenges to software engineering in this decade. We invited the authors of 59 of the 76 papers submitted to the workshop² to discuss the recent changes in software engineering, share a vision of the future evolution of the discipline, and shape a roadmap for the

research community. We invited the authors of selected papers to extend their paper for the SE2030 roadmap ACM TOSEM special issue, by taking into account the comments of the early reviews, and the outcome of the workshop that we summarize in this paper. The discussion identified seven key areas that challenge the research in software engineering:

Artificial Intelligence for Software Engineering (AI4SE) The recent breakthrough in machine learning, generative AI and autonomous systems triggers the deepest change in the skyline of software engineering research and practice since the Internet revolution in the second half of the last century [34] [4]. The software engineering community has never seen a so fast and predominant growth of new research threads such as the study of machine learning in software engineering and the challenges of engineering machine-learning-driven systems, both of which have become the dominant themes of the main software engineering conferences and journals [19]. The discussion in the workshop highlighted the challenges of AI tools to engineer software systems, the core challenges of predicting defects and trusting AI, and the deep need of reliable and explainable AI in the development workflow.

Software Engineering by and for Humans Machine learning, Generative AI and autonomous systems shape a new landscape for software engineering by and for humans, and radically change even the basic concept of software artifact [9]. The new software systems challenge software engineers with new ethical, technical and fairness problems [12]. Humans become integral part of large software ecosystems, and the research in software engineering shall move beyond the short-sighted vision of users of software systems, toward a new vision of humans as integral part of cyber-physical ecosystems [35]. The discussion in the workshop has focused on how LLMs bridge technical and human aspects, the ethical implications of AI, and the need for continuous human oversight in software development.

Sustainable Software Engineering The concept of sustainable development extends beyond classic environmental concerns, and spreads over software systems in cyber-physical spaces. Sustainable software operations in cyber-physical spaces require new design, development, deployment, and maintenance approaches that minimize the ecological footprint,

¹An excellent introduction to the liberating structure is available at <https://www.liberatingstructures.com>

²The list of workshop papers as well as the url of many of the papers is available at the workshop website <https://conf.researchr.org/home/2030-se>. Workshop pictures and latest updates are available at <https://www.inf.usi.ch/faculty/pezze/se2030.html>

enhance resource efficiency, and promote social responsibility [27]. The discussion in the workshop highlighted the complex intertwine of societal, environmental, and economic perspectives, the need for regulations and software sustainability measures, and the energy inefficiency of AI models.

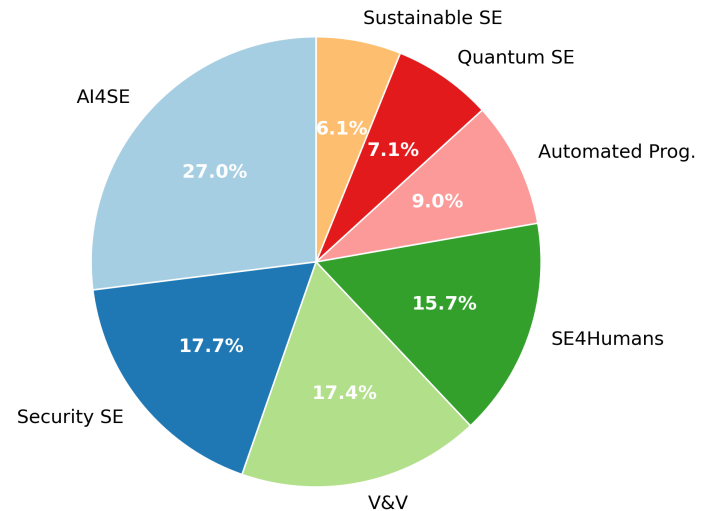
Automatic Programming Machine learning, and specifically deep neural networks and Large Language Models (LLMs), are the largest magnification factor of human productivity ever seen in software engineering since the pioneer times. They open new frontiers towards automatic programming, upset the quality and security scenario, and raise new societal and legal issues. The discussion in the workshop focused on the role of AI in generating code and upgrading legacy system [2], and in the challenges of context and prompt engineering.

Software Security Analysis The extensive usage of machine learning in software quality and security introduces new societal and legal considerations [1] [25]. As software systems grow in complexity and scale, new security challenges emerge, emphasizing the need for innovative approaches to secure software engineering and cybersecurity [16] [28]. The discussion in the workshop highlighted the importance of security-aware tools for vulnerability detection and secure code generation, and the role of deep learning in anomaly detection and secure software supply chains.

Validation and Verification Machine learning and AI upset the validation and verification scenario. On one side, the adaptive and evolving nature of AI-driven systems changes the concept itself of test input and oracle, on the other side machine learning and GAI open new frontiers to test automation. We need both a new conceptual approach to frame the problem of testing and analyzing AI-driven software systems and new studies on how exploit ML and GAI in software testing and analysis. The discussion in the workshop focused on the recent progress in knowledge compilation and meta-solvers that enhanced the scalability of various software engineering tools, and on the impact on test-case generation and configurable software analysis [21].

Quantum Software Engineering Quantum computing opens a new class of software systems that share very little if any at all with classic software systems. Quantum software engineering is still in its infancy and can greatly benefit from the open-minded studies of experienced software engineers towards new approaches to engineer quantum software systems. The discussion in the workshop focused on the challenges in quantum programming, including the lack of established knowledge, new design and testing methodologies, integration with classical systems, and the need for specialized education.

These research directions have significant implications for both academia and industry. Long term academic research shall develop new theories, methodologies, and educational curricula, to improve a deeper understanding of emerging technologies and their impacts. Industrial technology transfer shall use research results to design and implement innovative tools and practices that can improve software development, software quality and security, and address sustainability concerns. The following pie chart visualizes the trends of the discussion in the workshop, by showing the word count from the notes taken during the 2030 SOFTWARE ENGINEERING ROADMAP workshop:



Sections 2–8 briefly overview the discussion in the seven areas. Each section starts with a word cloud of the most frequent terms from the notes of the 2030 SOFTWARE ENGINEERING ROADMAP workshop, presents an overview of the main results, two lists of *open research questions* and *future research directions*, respectively, and conclude with an

Area summary.

Section 9 concludes with a summary of the main results of the 2030 SOFTWARE ENGINEERING ROADMAP workshop.

2. AI4SE



AI tools, like Copilot for code generation, are commonly integrated in industrial software development workflows [20]. The key open issues are (i) incorporating bug and vulnerability detection in the workflow, to enhance software quality and security, and (ii) guaranteeing trust in AI and explainability, especially in dependable or safety-critical systems. There is a need for AI development processes that ensure the reliability and understandability of AI-generated artifacts. We shall bridge software engineering and AI to seamlessly integrate requirements in the development of AI driven systems.

Open Research Issues

Explainability and Trust Ensuring that AI-generated outputs are explainable and trustworthy, particularly in safety-critical systems.

Quality Assurance Establishing quality measures for AI auto-coding to ensure that AI-generated code meets high standards.

Defect and Vulnerability Prediction Incorporating AI-based bug detection into the development workflow to enhance software quality and security.

Feedback Loops and Knowledge Integration Implementing feedback loops, using knowledge graphs, and enhancing more determinism in AI systems.

Testing and Security Developing hybrid testing approaches using AI and fuzzy techniques, and enforcing security guidelines in LLMs [17].

Domain Adaptation Applying deep learning and LLMs to new domains like quantum software engineering where data or well-defined knowledge may be sparse.

Human-AI Collaboration Enhancing AI training processes with human feedback, labeling, and crowdsourcing to improve model performance.

Future Research Directions

Bridging AI and SE Developing methodologies that integrate AI techniques into SE practices, focusing on sustainability, transparency, security, trustworthiness, and compliance [11].

Contextual Learning Incorporating context into AI training, as human learning heavily depends on contextual information.

Evaluating Costs Assessing the cost of training, fine-tuning, and using LLMs from different perspectives to determine the advantages of AI over other alternatives.

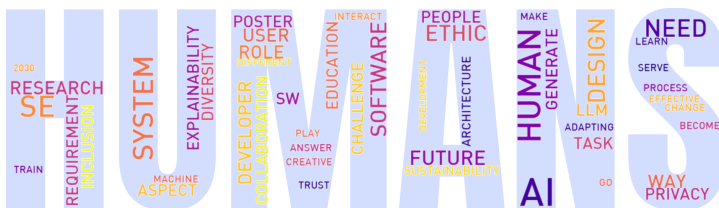
New Domains and Applications Exploring the use of AI in new domains such as quantum software engineering and low-resource environments.

Collaboration Between Academia and Industry Redefining the role of software engineers and adapting to the evolving landscape of AI and SE.

By addressing these open research issues and exploring future directions, the integration of AI in software engineering can be enhanced, leading to increasingly reliable, secure, and efficient software development processes.

AI4SE Future research should (i) develop robust quality measures for AI-generated code and integrate AI carefully into software development workflows and (ii) improve AI training with better feedback loops and multi-model approaches to address non-functional requirements.

3. SOFTWARE ENGINEERING BY AND FOR HUMANS



A core challenge that emerged in the 2030 SOFTWARE ENGINEERING ROADMAP workshop is the disconnection between software developers and users. Software engineers still build software for users according to some specifications. Popular software engineering approaches, like scrum, attempt to seamlessly include some users' needs in the development process with the role of product owner, however, there is still a large gap between users and software engineers. Large Language Models (LLMs) can bridge technical and human aspects [9]. Requirements engineering will play a significant role in instructing the machine to correctly capture the needs of the many individual users [30], with most activities being automated. AI can accelerate innovation, by radically changing the way humans interact with software systems. However, the role of humans in 2030 remains unclear. We need to understand which tasks will be performed by humans and which by AI. AI will dramatically change the way developers interact with systems. Keeping humans in the loop is essential to ensure meaningful engagement and development [34].

Open Research Issues

Ethical Rules and Regulations Defining and enforcing ethical rules and regulations, far beyond the AI Act and specific laws that have been recently approved.

Continuous Learning Mastering AI systems that continuously learn, thus impacting the way we generate code and potentially on human's creativity.

Explainability and User Adaptation LLM interfaces to let humans understand the decisions made in the background, and improve explainability. There is a concern that humans are adapting to systems rather than systems to humans, thus leading to a scenario where humans become passive entities.

Human-Centric Design Defining software development processes for humans, and balancing work-life and ethical issues [24].

Privacy and Regulations Rethinking both privacy by design and regulations on sustainability and AI in the software development process.

Diversity and Inclusion (DEI) Mastering the big AI systems' diversity and inclusion challenges [18], by both improving funding for DEI initiatives and addressing uncertainties about the impact on DEI research in academia.

Trust and Explainability Ensuring trust and understandability of AI systems.

Future Research Directions

AI and Human Collaboration Studying the potential of AI to improve both human-to-human and effective interactions [15].

Communication Channels Redefining communication channels between humans and AI systems as AI-powered tools, including AI-powered developers.

Privacy and Sustainability Defining software development paradigms and processes that implement privacy by design and adhere to stringent sustainability and AI regulations.

Automated Software Development Exploring design paradigms for fully AI-automated software development cycle from design to deployment and testing in fast sandbox environments.

User Involvement Ensuring that humans are first class actors in decision-making processes, and involving humans in system construction to build trustful systems [31].

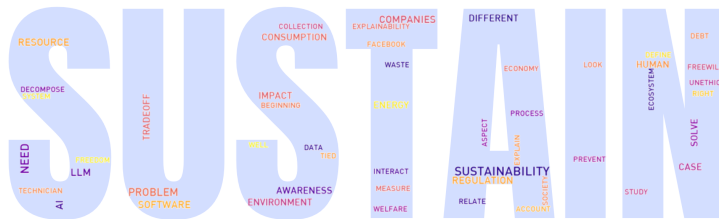
Transparency and Compliance Involving software engineering in implementing tools that provide insights on model training, enhancing transparency and compliance with upcoming regulations to democratize computer science and build trustful AI systems.

DEI Research Addressing the inconsistency in the quality of Diversity, Equity, and Inclusion (DEI) papers, and improving the reviewing processes to prevent the publication of misleading studies.

Education and Training Educating a new generation of software engineers, who can manage the new challenges of AI tools and software engineering [3].

Software Engineering by and for Humans: Research should focus on bridging the technical and human aspects of software engineering and defining ethical rules for AI systems. Improving human-to-human collaboration through AI and ensuring continuous human involvement are key to addressing ethical and interaction challenges.

4. SUSTAINABLE SOFTWARE ENGINEERING



Sustainable software engineering is becoming increasingly critical as we consider the welfare, freedom, and freewill within the human ecosystem. We need to approach sustainability from multiple perspectives: societal, environmental, and economic [6]. The importance of sustainability today is driven by the enormous demand for resources, evolving human values, and governmental regulations [13].

Open Research Issues

Corporate Responsibility Defining regulation to ensure that organizations adopt sustainable practices and prevent unethical data collection, as exemplified by cases like Meta³.

Energy Consumption Preventing useless usage of expensive and unnecessary AI. AI processing requires substantial energy, and LLMs are particularly energy inefficient [33]. Periodic sustainability benchmarks could help monitor and improve their impact. However, the negative impacts of LLMs on companies are still largely unknown. This ties to explainability that is a first required step to properly assign tasks to different entities.

³The New York Times, “Meta Fined \$1.3 Billion for Violating E.U. Data Privacy Rules, <https://www.nytimes.com/2023/05/22/business/meta-facebook-eu-privacy-fine.html>”

Metrics for Sustainability Measuring the multiple facets of sustainability from the inception of the projects. Software waste and technical debt are pressing issues. We need metrics of software sustainability. However the concept of metrics itself is ill-defined [14].

Awareness and Education Raising awareness about sustainability within companies, often unaware of the core issues of software sustainability. Education plays a crucial role in fostering a sustainable approach within the software engineering community [26] [32].

Trade-offs in Sustainability Assessing whether people are ready to make these trade-offs [10]. Sustainability often involves trade-offs, such as balancing perfect services against resource consumption.

Future Research Directions

Developing a Sustainability Mindset Studying and understanding the different dimensions of sustainability and developing a “sustainability mindset” within organizations. This involves raising awareness among researchers about sustainability.

Modulating Resource Consumption Automatically distributing the resource consumption of LLMs according to the nature of the problem.

Regulatory Compliance Ensuring compliance with regulations and laws that are increasingly tailored to software.

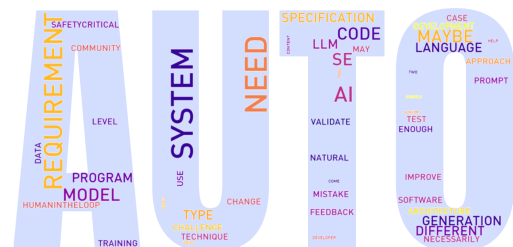
Developing Business Cases and Examples Developing business cases and case studies to witness the importance of sustainable practices.

Defining Sustainability Metrics and Benchmarks Defining sustainability metrics and benchmarks to reduce the environmental impact of software and AI systems. Establishing clear measures of software sustainability and integrating them from the beginning of a project is crucial for long-term success.

Promoting Educational Initiatives Developing educational initiatives to improve a sustainable approach within the software engineering community.

Sustainable Software Engineering: Research needs to develop specific sustainability metrics and foster a sustainability mindset within organizations. It is essential to balance resource consumption with service quality while minimizing software ecological footprint.

5. AUTOMATIC PROGRAMMING



Machine learning and generative AI already provide powerful tools for automatically generating code [23], and open important questions about the scalability of code generation as well as about quality and trustfulness of the generated code.

Open Research Issues

Program Synthesis Defining scalable ML-driven approaches to synthesize predictable, verifiable, and testable programs, by taking advantage of the results of template-based approaches.

Automation and Human-in-the-Loop Defining LLM-based agent systems to build teams and incorporate feedback. The degree of automation varies, and the human-in-the-loop issue may evolve over time. We need LLM-based agent systems to build teams and incorporate feedback.

Context and Adaptability Defining a comprehensive software engineering approach that encompasses many abstract levels such as requirements, architecture, specifications and validation, and is resistant to regulations, hardware, and architecture changes.

Validation and Formal Specifications Validating both system specifications and behavior through formal specifications, by transforming natural language into an intermediate level and then into code.

Domain-Specific Languages Defining domain-specific languages tailored to different types of systems, and integrated with LLMs for effective automatic programming.

Training Data and Model Performance Defining different models for various application domains, to ensure that training data do not lead AI to repeat human mistakes.

Human-in-the-Loop Interactions Keeping humans in the loop to refine and improve the outcomes of automatic programming that relies on the use of models[22].

Prototypes and Models Fast generating prototypes and models that encompasses code, as well as requirements, test cases, and design artifacts[5].

Conversational Approaches and Bug Handling Defining conversational approaches for improving requirements, and handling bugs, especially in safety-critical systems and non-functional requirements[36].

Training Data Quality Ensuring that training data do not lead AI to repeat human mistakes is vital. Different models may be required for various application domains.

Future Research Directions

Improving Program Synthesis Techniques Enhancing the longevity and effectiveness of program synthesis techniques.

Enhancing LLM Style Code Generation Developing robust prompt engineering methods, and integrating them into large systems.

Developing New Agent Systems Creating new agent systems based on LLMs that can build teams and incorporate feedback.

Context-Aware Systems Defining approaches to produce systems that adapt to changing contexts, such as regulations, hardware, and architecture.

Establishing Validation Methods Developing methods to validate system specifications and behavior through formal specifications.

Creating Domain-Specific Languages Defining domain-specific languages tailored to different systems, and integrating them with LLMs, to improve the effectiveness of automatic programming.

Ensuring High-Quality Training Data Ensuring that training data are diverse and accurate, and developing models for different application domains.

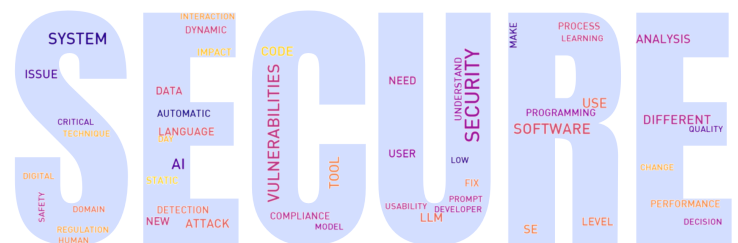
Enhancing Human-in-the-Loop Interactions Improving human-in-the-loop interactions to refine and improve AI outcomes.

Utilizing Prototypes and Models Emphasizing the use of prototypes and models in automatic programming.

Handling Bugs in Conversational Approaches Developing methods to handle bugs effectively in conversational approaches to improve the reliability of AI systems, especially in safety-critical applications.

Software Quality with Automatic Programming: Future work should (i) integrate code generation with Large Language Models (LLMs), (ii) improve prompt engineering techniques, (iii) develop domain-specific languages specific for different system types for improving software quality.

6. SOFTWARE SECURITY ANALYSIS



Software security analysis is crucial in today's landscape, where security threats and vulnerabilities are ever-present, and the domain language barrier complicates communication and tool integration, as terms have different meanings across various areas.

Open Research Issues

Domain Language Barrier Melting terms that are used with different meanings across areas by taking advantage of AI APIs to bridge the gaps.

Static and Dynamic Analysis Techniques Reducing the extensive validation and improving the usability of static and analysis techniques to find vulnerabilities.

Security-Aware Tools Defining security-aware tools that are pivotal in automatic vulnerability detection and compliance processes, such as risk modeling and minimum elements compliance.

Generating Secure Code Leveraging prompts and ensuring that LLMs contribute effectively, despite lower performance in dynamic analysis scenarios.

Detecting and Fixing Vulnerabilities Guaranteeing continuous commitment in vulnerability data and complexity in anomaly detection.

Deep Learning for Anomaly Detection Defining simple deep learning approaches to detect anomalies, and explain decisions.

Privacy Issues and Software Supply Chain Practices Enhancing API usability for developers with varying security expertise, and training developers with expertise in secure practices conforming to European Union data regulations.

AI Tools for Secure Algorithms Producing secure algorithms through a mix of experts, from natural language to machine instructions, requires tools for secure logic review [28].

Legal Accountability and Continuous Security Compliance Balancing legal and technical aspects for critical as well as non-critical systems, to both empower users and secure the interactions.

Edge Computation for Security Enhancing security by minimizing data exposure, vulnerabilities and securing system interactions, to prevent attacks to public institutes.

Learning Systems and AI in Security Defending against AI-based attacks and securing against vulnerabilities emphasizes the need for research in software security.

Future Research Directions

Addressing Domain Language Barriers Developing standardized terminology and AI APIs to improve communication and tool integration across different areas.

Improving Static and Dynamic Analysis Techniques Improving the usability and validation of tools to effectively uncover vulnerabilities.

Advancing Security-Aware Tools Creating effective automatic vulnerability detection tools and compliance processes, to both improve risk modeling and ensure compliance with security standards.

Enhancing Secure Code Generation Improving prompts and LLM performance in dynamic analysis scenarios, to generate more secure code.

Balancing Vulnerability Data Addressing the imbalance in vulnerability data, and simplifying anomaly detection methods, to improve detection and fixing processes.

Simplifying Deep Learning Anomaly Detection Developing deep learning-based anomaly detection that is both explainable and user-friendly.

Streamlining Security Issue Reporting Developing a responsive and efficient vulnerability disclosure process.

Strengthening Privacy and Supply Chain Security Enhancing secure software supply chain practices and complying with EU data regulations.

Developing AI Tools for Secure Algorithms Creating tools that combine expert knowledge and AI to produce and review secure algorithms.

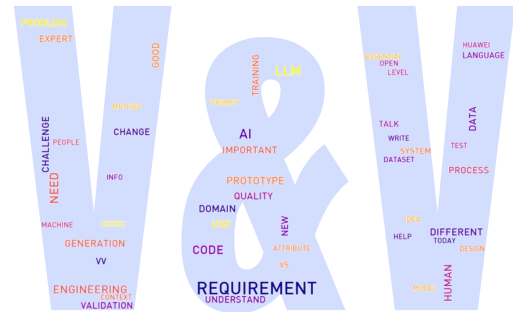
Ensuring Legal Accountability and Continuous Compliance Implementing continuous security compliance processes for both critical and non-critical systems.

Utilizing Edge Computation for Security Enhancing security by reducing data exposure and vulnerabilities, and addressing federation attacks.

Researching AI in Security Investigating the relationship between learning systems and AI to defend against AI-based attacks and secure systems.

Software Security Analysis: The research shall focus on (i) improving security-aware tools and anomaly detection with LLMs, (ii) improving the vulnerability disclosure process and (iii) securing software supply chains for robust software security.

7. VALIDATION AND VERIFICATION



Validating and verifying (V&V) code generated by machines poses unique challenges, and raises questions about how to assess quality and efficiency and ensure reliability of automatically generated software systems [8]. Requirements engineering in the context of AI systems demands transparency and new evaluation methods beyond traditional standards. Prompt engineering is crucial, as changing prompts can yield different outcomes from LLMs.

Open Research Issues

Quality and Efficiency Assessment Evaluating the quality, efficiency and reliability of machine-generated code.

Transparency in Requirements Engineering defining quality standards for machine-generated code, by incorporating transparency and trustfulness.

Impact of Prompt Engineering Defining consistent and effective prompt engineering strategies with predictable variability in the outcomes.

New Testing Methods Defining continuous validation approaches for evaluating machine-generated code [17].

Domain-Specific Needs Defining domain-specific methodologies to validate the requirements of different domains.

Robust Requirements Writing Writing robust quality requirements for automatically generate code, especially when automating code generation with LLMs.

Adapting Education in Requirement Engineering Training for requirement engineering by focusing on AI, automatic testing, and specialized processes.

Future Research Directions

Innovative Approaches to Prototyping Defining approaches for rapid prototyping from requirements, adapting to organizational contexts, and leveraging AI as a team member alongside domain experts to facilitate dialogue and ensure clarity [7].

Evolving Role of Requirement Engineers Improving the detection and refinement of requirements through collaborative tools and diverse data sets, with AI capabilities playing a significant role [30].

Human-Computer Interaction (HCI) in V&V Defining streamlining requirement processes with LLMs using HCI principles to ensure robust validation and verification.

Continuous Validation and Updates Establishing continuous validation processes to keep up with evolving requirements and AI capabilities.

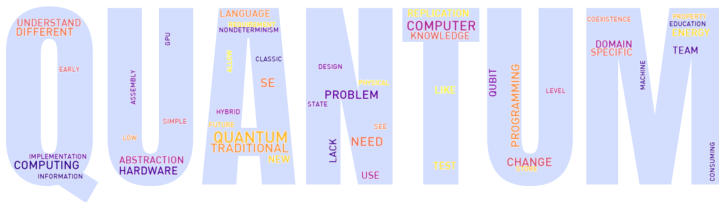
Improved Testing Methodologies Developing new testing methodologies that address the complexities of AI-generated code and the specific needs of various domains.

Integration of AI in Requirement Engineering Education Adapting educational programs to include AI-focused training for requirement engineering, emphasizing the importance of automatic testing and specialized processes.

AI-Assisted Requirement Refinement Utilizing AI to assist in the refinement and interpretation of requirements, ensuring accuracy and completeness.

Validation and Verification: Research should develop new evaluation methods and use AI to refine requirements and improve V&V transparency, by focusing on the improvement of human-computer interaction (HCI) in V&V processes for robust validation and verification of AI-generated code.

8. QUANTUM SOFTWARE ENGINEERING



Quantum Engineering raises unique challenges and requires deep changes in software engineering paradigms to cope with the distinctive characteristics of quantum computing. Requirements engineering remains fundamental, yet it may significantly differ from classic approaches. Programming in quantum environments resembles low-level assembly coding, and emphasizes the need for specialized education and training. Integration of quantum and classical computing systems may be possible with hybrid GPU setups.

Open Research Issues

Lack of Established Abstractions and Approaches Defining requirements engineering, adaptation, and design approaches that cope with the different abstractions in quantum programming.

Testing Methodologies Defining testing strategies for the non-replicable nature of qbit states of quantum software systems [29].

Non-Determinism and Compatibility Defining approaches that handle the non-determinism nature of quantum software systems and address the incompatibility with classic software architectures.

New Programming Paradigms Developing a mindset, programming paradigm, programming languages and methodologies for the distinctive nature of quantum engineering.

Energy-Intensive Hardware Balancing quantum and traditional computational resources to deal with the high energy costs of quantum computing.

Common Abstractions Developing common abstractions across quantum and traditional computing domains.

Specialized Education and Training Developing training material for educating and training quantum engineers and programmers.

Future Research Directions

Innovative Design and Requirements Engineering Exploring new design approaches and requirements engineering that accommodates the unique characteristics of quantum systems.

Testing and Verification Strategies Developing novel testing and verification strategies that account for the physical properties and non-deterministic nature of quantum computing [29].

Interdisciplinary Collaboration Leveraging interdisciplinary collaboration to navigate hardware constraints and address the unique demands of quantum software engineering.

Integration with Classical Systems Investigating the integration of quantum and classical computing systems to create hybrid solutions.

Education and Training Programs Establishing specialized education and training programs to equip software engineers with the necessary skills for quantum programming.

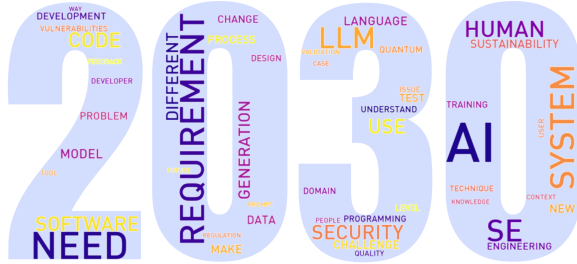
Energy Efficiency Researching ways to optimize energy consumption in quantum computing, balancing computation allocation between quantum and traditional hardware.

Development of New Languages Creating new programming languages and methodologies tailored to the unique requirements of quantum computing.

Common Abstractions and Frameworks Developing common abstractions and frameworks to bridge the gap between quantum and classical computing paradigms.

Quantum Software Engineering: Research should focus on (i) new design approaches and specialized training for quantum software to tackle unique quantum system challenges, and (ii) common abstractions for integrating quantum and classical computing systems.

9. FINAL CONSIDERATIONS



In the 2030 SOFTWARE ENGINEERING ROADMAP workshop we focus on seven core areas that deeply challenges the software engineering research and that we summarize in Table 1:

Artificial Intelligence for Software Engineering (AI4SE) The main challenges in AI4SE involve ensuring trust in AI-generated outputs and integrating AI tools into the software development workflow. Future research should focus on enhancing AI training processes, improving model performance through multi-model approaches and feedback loops, and addressing non-functional requirements.

Software Engineering by and for Humans The key challenge of AI-powered systems is the disconnection between software developers and users. Ethical implications and continuous human involvement are crucial. Future research directions include bridging the technical and human aspects of software engineering, defining ethical rules and regulations, and enhancing human-to-human collaboration facilitated by AI.

Sustainable Software Engineering The environmental impact of software, including the energy inefficiency of LLMs, and software waste are significant concerns. Future research should aim to develop sustainability benchmarks, build a sustainability mindset within organizations, and balance resource consumption with software quality.

Automatic Programming The core challenges of automatic programming include understanding the capabilities of models and developing new techniques for context and prompt engineering. Future research should focus on integrating code generation with LLMs, improving prompt engineering, and developing domain-specific languages and methodologies focused on automated programming.

Software Security Analysis Domain language barriers and usability challenges hinder effective security analysis. Future research directions include improving security-aware tools for automatic vulnerability detection, improving anomaly detection with deep learning, and developing secure software supply chain practices that comply with regulatory standards.

Validation and Verification (V&V) Validating and verifying machine-generated code presents unique challenges. Future research should develop new evaluation methods, leverage AI for requirement refinement, and improve human-computer interaction in V&V processes to ensure robust validation and verification.

Quantum Software Engineering Quantum software engineering is an emerging field, with significant knowledge gaps and non-determinism challenges. Future research should focus on developing new design approaches, providing specialized education and training in quantum programming, and exploring

common abstractions for integrating quantum and classical computing systems.

The *25/10 Crowd Sourcing*⁴ panel highlighted five top research challenges:

Challenge	SUM	MEAN
Evaluating the impact of future systems on society and sustainability	37	4.75
Measuring sustainability	34	4.40
Requirements for ML-enabled socio-technical systems	25	4.25
Cross-cutting requirements (security performance, sustainability)	25	4.17
Empowering the interactions between humans and self-adaptive autonomous systems	25	4.17

Impact of future systems on society and sustainability emerges as the most critical challenge with the highest sum of votes (37) and the highest mean score (4.75). This indicates a strong consensus on the importance of understanding how future technologies will affect societal and environmental aspects, reflecting a growing concern for sustainability and human-centric engineering.

Measuring sustainability is also highly rated, with a sum of 34 and a mean score of 4.40. This challenge highlights the need for effective metrics to quantify sustainability, which is crucial for making informed decisions about technology deployment and its long-term effects.

Requirements for ML-enabled socio-technical systems and **Cross-cutting requirements (security performance, sustainability)** both received the same cumulative rate (25) and mean score (4.17). The focus on machine learning-enabled systems and cross-cutting requirements suggests a significant interest in addressing the integration of advanced technologies with comprehensive criteria that include security and sustainability.

Approaches to empower users interactions with self-adaptive autonomous systems also scored a mean of 4.17. This challenge emphasized the need to design systems that both prioritize user needs and improve user experience, by ensuring that technology effectively serves the human needs.

The results clearly reflect the emphasis on the disruptive effects of AI, user-centric approaches and sustainability in the development of future software systems. These challenges reveal key areas where future research and development efforts should be concentrated to align technological advancements with societal needs and expectations.

Overall, the 2030 SOFTWARE ENGINEERING ROADMAP workshop has provided a comprehensive overview of the current and future trends in software engineering. The integration of AI, quantum computing, and autonomous systems into software engineering practices represents a paradigm shift, bringing significant advancements and posing new challenges. As the software engineering community addresses these changes, it is essential to focus

⁴An excellent introduction to the liberating structure is available at <https://www.liberatingstructures.com/12-2510-crowd-sourcing/>

Table 1: 2030 Open Issues and Future Research Directions in Software Engineering

Topic	Main Open Issues	Future Research Directions
AI for Software Engineering	Trust in machine-generated code, explainability, integration into workflows, quality measures for machine-generated code	Improving model performance with multi-models, defining feedback loops, addressing non-functional requirements, improving trainings
Software Engineering by and for Humans	Disconnection between developers and users, ethical implications, continuous human involvement	Bridging technical and human aspects, defining ethical rules and regulations, enhancing AI-human collaboration
Sustainable Software Engineering	Environmental impact, technical debt, energy inefficiency of LLMs	Developing sustainability metrics, building a sustainability mindset, balancing resource consumption with software quality
Automatic Programming	Capabilities of models, development of new techniques, context and prompt engineering	Integrating program synthesis and LLMs, improving prompt engineering, developing domain-specific languages
Software Security Analysis	Domain language barriers, usability challenges, continuous vulnerability detection	Improving security-aware tools, improving anomaly detection with deep learning, secure software supply chain practices
Validation and Verification (V&V)	Quality assessment of machine-generated code, transparency in requirements engineering, domain-specific needs	Developing new evaluation methods, leveraging AI for requirements refinement, enhancing HCI in V&V processes
Quantum Software Engineering	Lack of quantum physics knowledge, non-determinism, compatibility with classical computing	Developing new design approaches, specialized education and training, exploring common abstractions for quantum and classical systems

on ethical considerations, sustainability, security, and the evolving role of humans within software ecosystems. The collaboration between academia and industry will be crucial in redefining the role of software engineers and adapting to the rapidly evolving landscape of software engineering. By embracing these challenges and opportunities, the software engineering community can build a resilient and innovative future, ensuring that software systems remain reliable, secure, and aligned with societal values.

Acknowledgment

We would like to thank Daniel Russo for the invaluable help in defining the ideal liberation structure for the program of the workshop.

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Calendar of Future Events

For current information see:

SIGSOFT-sponsored conferences:

<http://www.acm.org/sigsoft/conferences/index.html> or

<http://campus.acm.org/calendar/index.cfm?Sponsor=SIGSOFT>

ACM-sponsored conferences: <http://campus.acm.org/calendar/>

IEEE-sponsored conferences:

<http://www.computer.org/conferences/calendar.htm>

To submit information to the **Calendar of Future Events**, send email to Calendar_SEN@sigsoft.org.

Conference	Support	2024
SOFLIM4KG 2024 – Baltimore	-	Nov
FASSI 2024 – Nice, France	-	3 – 7 Nov
FMAS 2024 – Manchester, UK	-	11 – 13 Nov
CASCON 2024 – Toronto, Canada	-	11 – 15 Nov
IFM 2024 – Manchester, UK	-	13 – 15 Nov
SCALABILITY 2024 – Valencia, Spain	-	17 – 21 Nov
EXPLAINABILITY 2024 – Valencia, Spain	-	17 – 21 Nov
VEHICULAR ANALYTICS 2024 – Valencia, Spain	-	17 – 21 Nov
SYNCHRON 2024 – Bamberg, Germany	-	18 – 22 Nov
CooPIS 2024 – Porto, Portugal	-	19 – 21 Nov
BENEVOL 2024 – Namur, Belgium	-	21 – 24 Nov
IMETA 2024 – Dubai, UAE	-	24 – 26 Nov
BCCA 2024 – Dubai, UAE	-	26 – 29 Nov
LADC 2024 – Recife - PE, Brazil	-	26 – 29 Nov
PROFES 2024 – Tartu, Estonia	-	2 – 4 Dec
MIDDLEWARE 2024 – Hong Kong, China	-	2 – 6 Dec
ICFEM 2024 – Hiroshima, Japan	-	2 – 6 Dec
QuASoQ 2024 – Chongqing, China	-	3 Dec
ICSOC 2024 – Tunis, Tunisia	-	3 – 6 Dec
APSEC 2024 – Chongqing, China	-	3 – 6 Dec
SBMF 2024 – Vitória - Espírito Santo, Brazil	-	4 – 6 Dec
SDS 2024 – Gran Canaria, Spain	-	9 – 11 Dec
PDCAT 2024 – Hong Kong, China	-	13 – 15 Dec
ICISS 2024 – Jaipur, India	-	16 – 20 Dec

Conference	Support	2025
HICSS 2025 – Big Island, Hawaii, USA	-	7 – 10 Jan
ICITS 2025 – Mexico City, Mexico	-	22 – 24 Jan
VAMOS 2025 – Rennes, France	-	4 – 5 Feb
CSL 2025 – Amsterdam, The Netherlands	-	10 – 14 Feb
ESOC 2025 – Bozen-Bolzano, Italy	-	20 – 21 Feb
ISEC 2025 – Kurukshetra, India	-	20 – 22 Feb
SRC 2025 – Las Vegas, USA	-	1 – 5 Mar
EuroSys 2025 – Rotterdam, The Netherlands	-	30 Mar – 3 Apr
SAC 2025 – Sicily, Italy	-	31 Mar – 4 Apr
EDCC 2025 – Lisbon, Portugal	-	8 – 11 Apr
ICSE 2025 – Ottawa, Canada	-	26 Apr – 4 May
ESOP 2025 – Hamilton, Canada	-	5 – 8 May

Conference	Support	2024
VALID 2024 – Venice, Italy	-	29 Sept – 3 Oct
SIMUL 2024 – Venice, Italy	-	29 Sept – 3 Oct
ICSEA 2024 – Venice, Italy	-	29 Sept – 3 Oct
ESWEEK 2024 – Raleigh, USA	-	29 Sept – 4 Oct
CBSOFT 2024 – Curitiba, Brazil	-	30 Sept – 4 Oct
LATAM School 2024 – Curitiba, Brazil	Sponsor	1 Oct
SERD 2024 – Oklahoma, USA	-	1 – 3 Oct
APM 2024 – Turin, Italy	-	2 – 4 Oct
MEMOCODE 2024 – Raleigh, USA	-	3 – 4 Oct
VISSOFT 2024 – Arizona, USA	-	6 – 7 Oct
SCAM 2024 – Arizona, USA	-	7 – 8 Oct
LPNMR 2024 – Texas, USA	-	11 – 14 Oct
ICLP 2024 – Texas, USA	-	11 – 17 Oct
PICS 2024 – Copenhagen, Denmark	-	14 – 18 Oct
FMCAD 2024 – Prague, Czech Republic	-	15 – 18 Oct
RTIS 2024 – My Tho, Vietnam	-	17 – 19 Oct
ISECON 2024 – Virtual	-	19 Oct
SAS 2024 – California, USA	-	20 – 22 Oct
SPLASH 2024 – California, USA	-	20 – 25 Oct
SLE 2024 – California, USA	-	20 – 25 Oct
ESEM/ESEIW 2024 – Barcelona, Spain	-	20 – 25 Oct
GPCE 2024 – California, USA	-	20 – 25 Oct
ATVA 2024 – Kyoto, Japan	-	21 – 24 Oct
APLAS 2024 – Kyoto, Japan	-	22 – 24 Oct
SEKE 2024 – San Francisco, USA	-	26 – 28 Oct
ASE 2024 – Sacramento, USA	-	27 Oct – 1 Nov
ECSS 2024 – Malta	-	28 – 30 Oct
ISSRE 2024 – Tsukuba, Japan	-	28 – 31 Oct
ER 2024 – Pittsburgh, USA	-	28 – 31 Oct
BAIQC 2024 – Roanne, France	-	30 – 31 Oct
ICTSS 2024 – London, UK	-	30 Oct – 1 Nov

Conference	Support	2025
CIBSE 2025 – Ciudad Real, Spain	-	12 – 16 May
EICS 2025 – Trier, Germany	-	24 – 28 Jun

*→SIGSOFT sponsored event;

→SIGSOFT in cooperation event

industry/commercial conferences

Upcoming Events

29 Sept – 3 Oct, VALID 2024: The Sixteenth International Conference on Advances in System Testing and Validation Lifecycle. Venice, Italy. Link: <https://www.iaria.org/conferences2024/VALID24.html>

29 Sept – 3 Oct, SIMUL 2024: The Sixteenth International Conference on Advances in System Simulation. Venice, Italy. Link: <https://www.iaria.org/conferences2024/SIMUL24.html>

29 Sept – 3 Oct, ICSEA 2024: The Nineteenth International Conference on Software Engineering Advances. Venice, Italy. Link: <https://www.iaria.org/conferences2024/ICSEA24.html>

29 Sept – 4 Oct, ESWEEK 2024: Embedded Systems Week is the premier event covering all aspects of hardware and software design for smart, intelligent and connected computing systems. This event brings together leading conferences, symposiums and workshops allowing attendees to benefit from a wide range of topics covering the state of the art in embedded systems research and development. Raleigh, North Carolina, USA. Link: <https://esweek.org/>

30 Sept – 4 Oct, CBSOFT 2024: The Brazilian Conference on Software: Practice and Theory (CBSOFT) is an event annually promoted by the Brazilian Computing Society (SBC) aiming at fostering the exchange of experience among researchers and practitioners from industry and academia about the most recent research, tendencies, and theoretical and practical innovations on software. Curitiba, Brazil. Link: <https://cbsoft.sbc.org.br/2024/cbsoft/>

*→ **1 Oct, LATAM School 2024:** The Fourth LATAM School in Software Engineering aims to help new and future Software Engineering researchers to be part of the larger software engineering community, launching a successful career and managing the challenges while feeling joy in their life as researchers. Curitiba, Brazil. Link: <http://cbsoft.sbc.org.br/2024/escola/>

1 – 3 Oct, SERD 2024: The Oklahoma International Publishing's (OkIP) International Conference on Software Engineering Research & Development (SERD). The conference aims to unite scholars from different disciplinary backgrounds and emphasize disseminating ongoing research and development. Oklahoma, USA. Link: <https://eventutor.com/event/54/>

2 – 4 Oct, APM 2024: The 6th International Asynchronous Programming Models Workshop is a venue for the discussion of

asynchronous programming patterns, languages and model, ranging from theoretical results, over applications and implementations to opinion and overview contributions. Turin, Italy. Link: https://edkamb.github.io/APM_24/

3 – 4 Oct, MEMOCODE 2024: The 22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design. Raleigh, USA. Link: <https://memocode2024.github.io/>

6 – 7 Oct, VISSOFT 2024: The 12th IEEE Working Conference on Software Visualization. Arizona, USA. Link: <https://vissoft.info/2024/>

7 – 8 Oct, SCAM 2024: The 24th IEEE International Conference on Source Code Analysis and Manipulation (SCAM) aims to bring together researchers and practitioners working on theory, techniques, and applications that concern analysis and/or manipulation of the source code of software systems. Arizona, USA. Link: <https://conf.researchr.org/home/scam-2024>

11 – 14 Oct, LPNMR 2024: The 17th International Conference on Logic Programming and Non-monotonic Reasoning. Dallas, Texas, USA. Link: <https://lpnmr2024.demaccs.unical.it/>

11 – 17 Oct, ICLP 2024: The 40th International Conference on Logic Programming. Dallas, Texas, USA. Link: <https://www.iclp24.utdallas.edu/>

14 – 18 Oct, PICS 2024: The 5th Ph.D. school in the series of Foundations of Programming and Software Systems. Copenhagen, Denmark. Link: <https://etaps.org/about/fopss-schools/2024/>

15 – 18 Oct, FMCAD 2024: Formal Methods in Computer-Aided Design. Prague, Czech Republic. Link: <https://fmcad.org/FMCAD24/>

17 – 19 Oct, RTIS 2024: The Sixth International Conference on Real-time Intelligent Systems. My Tho, Vietnam. Link (insecure): <https://socio.org.uk/rtis/>

19 Oct, ISECON 2024: The Information Systems Education Conference provides a wonderful opportunity for CIS, MIS, IT, and CS faculty members from a wide variety of institutions to gather, share experiences, and learn from each other. Virtual. Link: <https://www.isecon.org/>

20 – 22 Oct, SAS 2024: The 31st Static Analysis Symposium. Pasadena, California, USA. Link: <https://2024.splashcon.org/home/sas-2024>

20 – 25 Oct, SPLASH 2024: The ACM Conference on Systems, Programming, Languages, and Applications: Software for Humanity. Pasadena, California, USA. Link: <https://2024.splashcon.org/>

20 – 25 Oct, SLE 2024: The 17th ACM SIGPLAN International Conference on Software Language Engineering. Pasadena, California, USA. Link: <https://www.sleconf.org/2024/>

20 – 25 Oct, ESEM/ESEIW 2024: The Empirical Software Engineering International Week and the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement. Barcelona, Spain. Link: <https://conf.researchr.org/home/esem-2024>

20 – 25 Oct, GPCE 2024: The ACM SIGPLAN International Conference on Generative Programming: Concepts & Experiences (GPCE) is a programming languages conference focusing on techniques and tools for code generation, language implementation, and product-line development. California, USA. Link: <https://2024.splashcon.org/home/gpce-2024>

21 – 24 Oct, ATVA 2024: The 22nd International Symposium on Automated Technology for Verification and Analysis. Kyoto, Japan. Link: <https://atva-conference.org/2024/>

22 – 24 Oct, APLAS 2024: The 22nd Asian Symposium on Programming Languages and Systems. Kyoto, Japan. Link: <https://conf.researchr.org/home/aplas-2024/>

26 – 28 Oct, SEKE 2024: The 36th International Conference on Software Engineering and Knowledge Engineering. San Francisco, USA. Link: <https://ksiresearch.org/seke/seke24.html>

27 Oct – 1 Nov, ASE 2024: The 39th IEEE/ACM International Conference on Automated Software Engineering. Sacramento, California, USA. Link: <https://conf.researchr.org/home/ase-2024>

28 – 30 Oct, ECSS 2024: The 20th anniversary celebrations of European Informatics Leaders Summit. Maltal. Link: <https://www.informatics-europe.org/ecss/home.html>

28 – 31 Oct, ISSRE 2024: The 35th IEEE International Symposium on Software Reliability Engineering. Tsukuba, Japan. Link: <https://issre.github.io/2024/>

28 – 31 Oct, ER 2024: The 43rd International Conference on Conceptual Modeling with theme “Conceptual Modeling, AI, and Beyond”. Pittsburgh, USA. Link: <https://sei.cmu.edu/go/er2024>

30 – 31 Oct, BAIQC 2024: The International Conference on Artificial Intelligence and Quantum Computing. Roanne, France. Link: <https://confscience.com/baiqc/>

30 Oct – 1 Nov, ICTSS 2024: The 36th International Conference on Testing Software and Systems. London, UK. Link: <https://conf.researchr.org/home/ictss-2024>

Nov, SOFLIM4KG 2024: The Software Lifecycle Management for Knowledge Graphs Workshop. Baltimore, USA. Link: <https://dachafra.github.io/soflim4kg/>

3 – 7 Nov, FASSI 2024: The Tenth International Conference on Fundamentals and Advances in Software Systems Integration. Nice, France. Link: <https://www.iaria.org/conferences2024/FASSI24.html>

11 – 13 Nov, FMAS 2024: The Sixth Workshop on Formal Methods for Autonomous Systems. FMAS is an international

workshop that brings together researchers working on a range of techniques for the formal verification of autonomous systems, to present recent work in the area, discuss key challenges, and stimulate collaboration between autonomous systems and formal methods researchers. Manchester, UK. Link: <https://fmasworkshop.github.io/FMAS2024/>

11 – 15 Nov, CASCON 2024: The 34th International Conference on Collaborative Advances in Software and Computing. Toronto, Canada. Link: <https://cascon.ca/>

13 – 15 Nov, iFM 2024: 19th International Conference on Integrated Formal Methods. Manchester, UK. Link: <https://ifm2024.cs.manchester.ac.uk>

17 – 21 Nov, SCALABILITY 2024: The First International Conference on Systems Scalability and Expandability. Valencia, Spain. Link: <https://www.iaria.org/conferences2024/SCALABILITY24.html>

17 – 21 Nov, EXPLAINABILITY 2024: The First International Conference on Systems Explainability. Valencia, Spain. Link: <https://www.iaria.org/conferences2024/EXPLAINABILITY24.html>

17 – 21 Nov, VEHICULAR ANALYTICS 2024: The First International Conference on Vehicular Systems inaugurates a series of events dedicated to analyzing solutions and challenges concerning dedicated frameworks, architectures, applications, software/hardware, tools and apps, experimental systems, and lessons learned in various aspects of vehicular systems. Valencia, Spain. Link: <https://www.iaria.org/conferences2024/VEHICULARANALYTICS24.html>

18 – 22 Nov, SYNCHRON 2024: The 31st International Open Seminar on Synchronous Programming is the place for lively discussions and presentations on Synchronous Languages and related topics. Bamberg, Germany. Link: <https://www.uni-bamberg.de/gdi/synchron-2024/>

19 – 21 Nov, CoopIS 2024: The 30th International Conference on Cooperative Information Systems is an established international event for presenting and discussing scientific contributions about technical, economical, and societal aspects of distributed information systems at scale. Porto, Portugal. Link: <https://coopis.scitevents.org/>

21 – 24 Nov, BENEVOL 2024: The 23rd Belgium-Netherlands Software Evolution Workshop o bring together researchers who are working in the field of software evolution and maintenance. Namur, Belgium. Link: <https://benevol2024.github.io/>

24 – 26 Nov, iMETA 2024: The 2nd International Conference on Intelligent Metaverse Technologies & Applications. Dubai, UAE. Link: <https://imeta-conference.org/>

26 – 29 Nov, BCCA 2024: The 6th International Conference on Blockchain Computing and Applications. Dubai, UAE. Link: <https://bccca-conference.org/2024/index.php>

26 – 29 Nov, LADC 2024: The 13th Latin-American Symposium on Dependable and Secure Computing. Recife - PE, Brazil. Link: <https://ladc.sbc.org.br/2024/>

2 – 4 Dec, PROFES 2024: The 25th International Conference on Product-Focused Software Process Improvement. Tartu, Estonia. Link: <https://conf.researchr.org/home/profes-2024>

2 – 6 Dec, Middleware 2024: The 25th ACM/IFIP International Middleware Conference is a major forum for discussing innovations and recent scientific advances in middleware systems with a focus on the design, implementation, deployment, and evaluation of distributed systems, platforms, and architectures for computing, storage, and communication. Hong Kong, China. Link: <https://middleware-conf.github.io/2024/>

2 – 6 Dec, ICFEM 2024: The 25th International Conference on Formal Engineering Methods. Hiroshima, Japan. Link: <https://icfem2024.info>

3 Dec, QuASoQ 2024: The 12th International Workshop on Quantitative Approaches to Software Quality. Chongqing, China. Link: <https://quasq2024.swc-rwth.de/>

3 – 6 Dec, ICSOC 2024: The 22st International Conference on Service-Oriented Computing. Tunis, Tunisia. Link: <https://icsoc2024.redcad.tn/>

3 – 6 Dec, APSEC 2024: The 31st Asia-Pacific Software Engineering Conference. Chongqing, China. Link: <https://conf.researchr.org/home/apsec-2024>

4 – 6 Dec, SBMF 2024: The 27th Brazilian Symposium on Formal Methods. Vitória - Espírito Santo, Brazil. Link: <https://sbmf24.ifes.edu.br/>

9 – 11 Dec, SDS 2024: The 11th International Conference on Software Defined Systems. Gran Canaria, Spain. Link: <https://emergingtechnet.org/SDS2024/>

13 – 15 Dec, PDCAT 2024: The 25th International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT). Hong Kong, China. Link: <https://hpcc.siat.ac.cn/meeting/pdcat2024/index.html>

16 – 20 Dec, ICISS 2024: The 20th International Conference on Information Systems Security. Jaipur, India. Link: <https://iciss.isrde.in/>

7 – 10 Jan, HICSS 2025: The Hawaii International Conference on System Sciences. Big Island, Hawaii, USA. Link: <https://hicss.hawaii.edu/>

22 – 24 Jan, ICITS 2025: The 8th International Conference on Information Technology & Systems is an international forum for researchers and practitioners to present and discuss the most recent innovations, trends, results, experiences and concerns in the several perspectives of Information Technology & Systems. Mexico City, Mexico. Link: <https://icits.me/>

4 – 6 Feb, VAMOS 2025: The 19th International Working Conference on Variability Modelling of Software-Intensive Systems. Rennes, France. Link: <https://familiar-project.github.io/VaMoS2025/>

10 – 14 Feb, CSL 2025: The 33rd Annual Conference on Computer Science Logic. Amsterdam, The Netherlands. Link: <https://cs12025.github.io/>

20 – 21 Feb, ESOC 2025: The 12th European Conference On Service-Oriented And Cloud Computing is the premier conference on advances in the state-of-the-art and practice of Service-Oriented Computing and Cloud Computing in Europe. Bozen-Bolzano, Italy. Link: <https://conf.researchr.org/home/esoc-2025>

20 – 22 Feb, ISEC 2025: The Innovations in Software Engineering Conference, ISEC, is a premier event dedicated to exploring the latest advancements, trends, and challenges in the field of software innovation. NIT Kurukshetra, India. Link: <https://conf.researchr.org/home/isec-2025>

1 – 5 Mar, SRC 2025: The ACM Student Research Competition call at the International Symposium on Code Generation and Optimization. Las Vegas, USA. Link: <https://2025.cgo.org/track/cgo-2025-student-research-competition>

30 Mar – 3 Apr, EuroSys 2025: The 20th annual European Conference on Computer Systems is a premier international forum for presenting computer systems research. Rotterdam, The Netherlands. Link: <https://eurosys25.hotcrp.com/>

31 Mar – 4 Apr, SAC 2025: The 40th ACM/SIGAPP Symposium on Applied Computing. Sicily, Italy. Link: <https://www.sigapp.org/sac/sac2025/>

8 – 11 Apr, EDCC 2025: The 20th European Dependable Computing Conference. Lisbon, Portugal. Link: <https://edcc2025.campus.ciencias.ulisboa.pt/>

26 Apr – 4 May, ICSE 2025: The International Conference on Software Engineering (ICSE) is the premier forum for presenting and discussing the most recent and significant technical research contributions in the field of Software Engineering. Ottawa, Canada. Link: <https://conf.researchr.org/home/icse-2025>

5 – 8 May, ESOP 2025: The 34rd European Symposium on Programming. Hamilton, Canada. Link (from search): <https://etaps.org/2025/conferences/esop/>

12 – 16 May, CIBSE 2025: The 28th Ibero-American Conference on Software Engineering aims to promote high quality scientific research in Iberoamerican countries and, thus, to support researchers in this community in the publication and discussion of their work. Ciudad Real, Spain. Link: <https://conf.researchr.org/home/cibse-2025>

24 – 28 Jun, EICS 2025: The 17th ACM SIGCHI Symposium on Engineering Interactive Computing Systems. EICS is devoted

to engineering interactive computing systems and their user interfaces, addressing one or more software quality factors, such as usability, user experience, reliability, security, etc. Trier, Germany. Link: <https://eics.acm.org/2025/>

Calls for Papers

SCIENCE DIRECT 2024: Information and Software Technology journal's special issue on Agile and Lean: How far did we come and what's next? This special issue aims to reflect on the progress in agile and lean practices and envision what future research and practice is needed in the agile and lean community. This special issue calls for submissions addressing topics across the full spectrum of agile and lean software development, broadly on agile, and on specific issues of concern to researchers or practitioners (or both). Papers should focus on the following themes: 1) Empirical studies and evaluations of the progress in the field 2) Foundations and conceptual studies 3) Evaluation-based research agendas 4) Evolution of agile practices and agile culture In the context of these themes, the topics of interest include: a) Agile leadership and culture b) Business agility and Startups c) Engineering (software design, architecture, DevOps, automation) d) Data-driven agility and transformations e) Agile product design and requirements. **Deadline:** Submission **11 October 2024**. CFP: <https://www.sciencedirect.com/journal/information-and-software-technology/about/call-for-papers#agile-and-lean-how-far-did-we-come-and-what-s-next>

EMSE 2024: Empirical Software Engineering journal's Special Issue on: Security Testing for Complex Software Systems (SECUTE). Software security and security testing are crucial for practitioners who must design and implement systems to minimize incidents such as data breaches, which can lead to severe consequences, including technical infrastructure damage, reputation loss, and legal issues. This special issue aims to offer practitioners and researchers insights, innovations, and solutions in security testing, focusing on complex and non-traditional software domains like AI, IoT, and cyber-physical systems. Topics of interest include, but are not limited to, the following: 1) Studies on the effectiveness of security testing in complex architectures and systems, e.g., AI-enabled, cyber-physical, IoT, and Virtual Reality systems 2) Adoption of security testing in non-source artifacts, like Infrastructure-as-Code scripts 3) Empirical studies on security testing methodologies 4) Presentation of novel methods for soliciting the adoption of security testing during the development. 5) Presentation of novel automated tools for security testing 6) Evolution and improvement of existing methods and techniques for security testing 7) Assessment or re-evaluation of existing automated tools for security testing 8) User studies to understand the experience of software engineers working with security testing 9) Case studies on real-world contexts where security testing practices are adopted. 10) Lessons learned and challenges faced while dealing with security in real-world complex systems. **Deadline:** Submission **31 October 2024**. CFP: https://emsejournal.github.io/special_issues/2024_SI_SECUTE.html

IEEE Software 2025: Special Issue on Quantum Software and

its Engineering. This special issue focuses on the emerging topic of developing software to solve real life problems using quantum computers. This issue aims to highlight current research in this direction and share experience of key players of the field at the current state-of-the-art. We invite papers covering any aspect of Quantum Software and its Engineering including, but not limited to: 1) Compilation of quantum algorithms, circuit synthesis and optimization 2) Tools for quantum software development 3) Programming languages for quantum computers 4) Test methods including software verification and validation, as well as quantum software quality in general 5) Empirical studies on quantum software engineering 6) Benchmarking quantum computing with software and noise models 7) Various aspects of quantum software engineering, at all phases of the software lifecycle like analysis, design, implementation, testing, deployment, or maintenance 8) Early applications of quantum error correction, quantum error mitigation 9) Software for noisy intermediate-scale quantum (NISQ) computers 10) Experience reports on current challenges and limitations. For a full list of topics please see CFP link. **Deadline:** Submission **10 October 2024**. CFP: <https://www.computer.org/digital-library/magazines/so/cfp-quantum-software-and-engineering>

ESOP 2025: The European Symposium on Programming is an annual conference devoted to fundamental issues in the specification, design, analysis, and implementation of programming languages and systems. ESOP seeks contributions on all aspects of programming language research including, but not limited to, the following areas: 1) Programming paradigms and styles 2) Methods and tools to specify and reason about programs and languages 3) Programming language foundations 4) Methods and tools for implementation, concurrency and distribution 5) Applications and emerging topics. Contributions bridging the gap between theory and practice are particularly welcome. 5 - 8 May 2025. Hamilton, Canada. **Deadline:** Round 2 Submission **10 October 2024**. CFP: <https://etaps.org/2025/conferences/esop/>

VAMOS 2025: The 19th International Working Conference on Variability Modelling of Software-Intensive Systems aims to bring together researchers and practitioners to share ideas, results, and experiences about their quest for mastering variability. VaMoS is a venue to discuss breakthroughs and experiences — academic or industrial — about variability-aware modelling techniques and their applications to software-intensive systems. Topics of interest include, but are not limited to: 1) Variability Modeling and Realization 2) Variability in Emerging Domains/Technologies 3) Automated Reasoning on Variability 4) Variability in AI Methods 5) AI for Variability (Generative AI, Large Language Models) 6) Configuration Management 7) Evolution of Variability-Intensive Systems 8) Runtime Variability 9) Variability and quality requirements 10) Industrial development of variable systems 11) Experience reports from managing variability in practice. For a full list of topics please see CFP link. 4 - 6 February 2025. Rennes, France. **Deadline:** Abstract **25 October 2024** Paper **1 November 2024**. CFP: <https://familiar-project.github.io/VaMoS2025/cfp/>

SRC at CGO 2025: The ACM Student Research Competition call at the International Symposium on Code Generation and Optimization (CGO). The competition offers a unique forum for

undergraduate and graduate students to present their original research before a panel of judges and attendees at CGO. To participate in the competition, a student must submit an extended abstract (500 words). Submissions in the form of an extended abstract are solicited in any topics relevant to the main conference, including: 1) Code Generation, Translation, Transformation, and Optimization for performance, energy, virtualization, portability, security, or reliability concerns, and architectural support 2) Efficient execution of dynamically typed and higher-level languages Optimization and code generation for emerging programming models, platforms, domain-specific languages 3) Dynamic/static, profile-guided, feedback-directed, and machine learning-based optimization 4) Program characterization methods 5) Compiler design, practice, and experience 6) Compiler abstraction and intermediate representations 7) Parallelism, heterogeneity, and reconfigurable architectures 8) Compiler-support for vectorization, thread extraction, task scheduling, speculation, transaction, memory management, data distribution, and synchronization. For a full list of topics and competition details please see CFP link. 1 - 5 March 2025. Las Vegas, USA. **Deadline:** Submission **15 December 2024**. CFP: <https://2025.cgo.org/track/cgo-2025-student-research-competition>

CibSE 2025: The 28th Ibero-American Conference on Software Engineering aims to promote high quality scientific research in Iberoamerican countries and, thus, to support researchers in this community in the publication and discussion of their work. In addition, the conference fosters collaboration and scientific production between academics, students and practitioners. Topics of interest include, but are not limited to: 1) Application of Artificial Intelligence to Software Engineering 2) Community-based software engineering (e.g., open source, crowdsourcing) 3) LLMs and other foundation models in Software Engineering 4) Open Science for Software Engineering 5) Software economics 6) Software evolution and modernisation 7) Software processes 8) Software reliability 9) Software repository mining and software analytics. For a full list of topics please see CFP link. 12-16 May 2025, Ciudad Real, Spain. **Deadline:** Abstract **17 January 2025** Paper **31 January 2025**. CFP: <https://conf.researchr.org/track/cibse-2025/cibse-2025>

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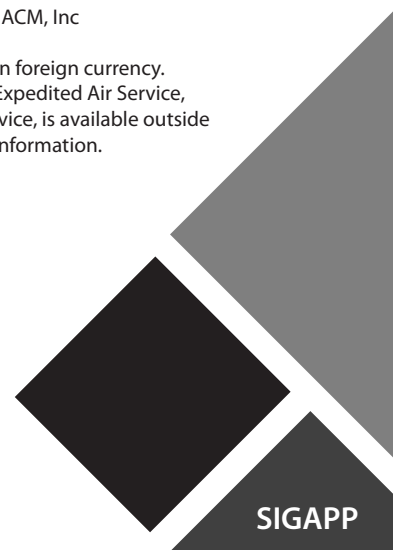
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